भारतीय प्रबंध संस्थान बेंगलूर
INDIAN INSTITUTE OF MANAGEMENT
BANGALORE

COMPLETE GUIDE TO PRODUCT MANAGEMENT

A guide on how to prepare for the Product Management Companies that come to IIM Bangalore

Academic Calendar: 2022-2023

SIGMA
PM & TECH CLUB

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Introduction

Hello Dear Reader!

Welcome to this modest attempt at passing on some knowledge, or Gyan as you might call it. The purpose of writing this book is to give you insights into acing the 'Product Management' interviews for companies that visit IIMB during the summer-placement internship process.

Some of you might be wondering, 'What does Product Management mean?' That's ok. Many of us didn't know either what Product Management (called PM from now on) meant and how to go about preparing for the role. We learnt along the way, a lot from material available on the internet and a lot from seniors in our time.

'You mentioned material. If there are resources out there already, why do we need this book?'

Good question. The simple answer to that is, while we were preparing for the PM role, we got a lot of unorganized, and sometimes, confusing advice from a lot of sources. Some sources asked us to do something while other sources asked us to do something else. We felt that there was a need for some standardization of the knowledge that was being passed on. Thus, this book is an attempt at standardizing some of the advice and knowledge that we received throughout our own prep and convey it in a way that makes prepping for the PM role more structured.

Book Structure

This book is having the theoretical content of the various topics that you need to prepare for cracking the interview. Some pointers about the book:

1. Each chapter has a couple of practice questions where we walk you through each step on how to go about solving them.
2. Under solved examples section we have attached video links of potential questions that you could refer in each chapter to get a better grasp of your concepts.
3. There are interview experiences of people in the summers for the last 5 years attached separately to this book.

The chapters in the book are divided as following:

1. **Ch 1&2:** What is Product Management? How to prepare for it?
2. **Preclude to Chapter 3:** Introducing you to Core concept of Product Management (A Must Read)
3. **Ch 3 - 10:** Core Concepts – Product Design, Feature Prioritization, Metrices, Guesstimates, RCA, App Critique, and Dashboards, Behavioral Questions (A Must Prepare)
4. **Ch 11 – 15:** Advanced Topics – Project Execution, System Design, Company Frameworks, Wireframing and Product Strategy (Optional)
5. **Conclusion**
6. **Resources and Cheat Sheets** (A Must Read)

How To Read This Book?

1. This book introduces you to the topic and subtopics. The idea of this book is to provide you an exhaustive list of topics for you to prepare for and get you acquainted on the type of questions usually asked.

2. Please go through one chapter a day, do not be in a haste to prepare all the chapters in limited time.

3. While reading through the chapters, understand the core concepts. In case you have any doubts please look at the "**Solved Problems**" section to understand with the help of a video. We'd suggest you solve them by yourself first before watching the video. You should try to identify where and how did you differ form an answer and analyze to see you have not missed any obvious things and learn from your shortcomings.

4. Try to read the Practice examples under each chapter **twice,** once to understand how to approach such a question and getting the structure right and the second time on how you would have solved this differently. Remember there are multiple solutions or ways to solve the same problem. *Quality of Preparation and not quantity matters!!*

5. Find focused people who want to target PM and discuss the cases in this book in depth with them to understand multiple viewpoints and approaches.

6. *The adjacent books which talk about interview experiences are tagged with which topics does a particular question asked in the interview refers to*. This is for you to know if a particular interview has the questions on the topic you studied for. Please check out the keyword **"PM Topics Covered".**

7. **Solve the interview experiences in a group preferably and brainstorm on different approaches.**

Disclaimers

- The methods discussed in the book are just one of the many ways to solve the problem and should be treated as a mere guidance rather than the exact right answer.
- Reading the book doesn't guarantee you of getting that PM internship but use it as a resource to guide you with how to get there and practice the things written here to prepare for the questions and cases and might come on the GD/Interview Day.

Let's begin this journey of preparing for a PM role.

Don't forget to, **Work Hard! Play Hard! Stay Motivated!**

Chapter 1: Introduction to Product Management

What is Product Management?

If you try finding the answer to this question on the internet or ask people around, you'd get the simplest and yet most confusing answer "CEO of the product". Now this paves the way for the next question, "What does a CEO actually do?". So instead of making things more complicated, we'll try to answer this question by breaking it into two parts:

1. **What is a product?** Goods or services that a company sells
2. **What is management?** Planning and organizing activities to achieve a specific goal

By combining these two definitions, we can define **Product Management** as-
Planning and organizing various tasks to ensure a product serves its intended function to the consumers throughout its lifespan.

The multiple tasks a Product Manager is responsible for include:

1. Designing the features of the product
2. Designing and building a product prototype
3. Testing the prototype
4. Making the required changes to the product prototype based on user feedback and usability
5. Designing and developing the final product
6. Maintaining the product

In conclusion, a Product Manager oversees every stage of the product life cycle, from design to maintenance.

Which skills are required to be a good Product Manager?

While we are still attempting to come up with a solution to this answer, we can direct you to the basic abilities that a PM uses daily. It is clear from looking at the various stages of the product lifecycle that there is a lot of decision-making, weighing options, and planning involved in answering questions like what features the product should have, how the prototype should be constructed, what metrics should be considered to measure the product's success, etc.
This clearly suggests that a PM must be

1. **Very organized in their thinking** and
2. Be able to use a **step-by-step approach** to decision-making.
3. The product's success ultimately depends on how consumers perceive it; thus, the PM must maintain constant communication with potential customers to understand their needs and pain points. Therefore, a PM must have effective **communication skills**.
4. Any company's aim is ultimately profits, so resource allocation is one of the most important tasks for a PM. Hence, it is important to come up with the best possible product/service in

the shortest time and with a well-defined budget. Therefore, a PM must have **critical thinking** to evaluate choices and make decisions within limitations.

5. Finally, a PM must convince clients of the product's earnest utility, which necessitates decent **marketing abilities**.

Types of Product Managers

Although this section may not seem very pertinent from the perspective of internships, it will give you an understanding of the many kinds of responsibilities that a PM has to perform depending on the job role. Additionally, we have mentioned a few companies (just for reference) with their respective roles. Three main verticals have been used to divide the sorts of PM:

1. Technology
2. Business
3. Design

Technology

Technical Product Managers

Their job includes researching and studying rival goods and services, working with development leads, offering product training, and analyzing market trends. (ADP, Amazon, HackerRank)

Digital Product Managers

The development and design of digital goods are within DPM's control. Then, they transform those demands into features that their team will design. They investigate digital goods and their features, organize product launch campaigns, and get in touch with customers to learn more about the wants and requirements of the market. (IndusInd, Gartner, Wells Fargo, Goldman Sachs)

Software Product Managers

They identify client needs and collaborate with the design team to develop features and products that address them. They also identify the software solutions that are out of date, and which should be phased away. (Lenovo, Logitech)

Business

Product Operations Managers

They carry out best practices, encourage resource sharing, and support ongoing training. They also strengthen departmental communication channels. (Uber, PayTM, Google)

Data Product Managers

They look for methods to make the most of data rather of merely using it. They also use product data for early design iterations and design enhancements. (Michelin, Airtel)

Product Marketing Managers

They are technical experts who know how to promote the company's products. Their job is to understand the utility of the features from consumers' perspective and market the product in the best relatable manner. (PhonePay, Amazon, Xiaomi)

Growth Product Managers
When it comes to experimenting with product design and ideas for new features in already-existing goods, GPMs take the lead. Additionally, they produce data-driven decision-making techniques. (Meesho, Uber, Unilever)

Ecommerce Product Managers
They oversee improving the performance and sales of the company's products across all e-commerce platforms. (WakeFit, Google)

Design

Product Design Managers
They coordinate the task of developing new product features and enhance existing ones while assuring design viability. (PayTM, RazorPay, PhonePay)

Types of products a PM might encounter

There are generally seven different types of products a PM encounters within his career:

SaaS or B2B Products
They are geared at meeting the demands of other businesses. They majorly address the financial and operational needs of the clients. Example: Salesforce

Mass Consumer Apps
They are designed to satisfy the straightforward needs of the mass market. Making the product appealing to the masses while being functional at the same time is the main goal. Example: Instagram and Uber

Additive Features:
They are a single component of a whole product that has the potential to have the main USP. The goal is to improve the product's overall functionality. Example: The shopping cart function on ecommerce

Early-Stage Products:
These are what today's entrepreneurs refer to as the Most Viable Product (MVP). As further development occurs after launch, depending on user input and product performance analysis, the emphasis is on quick development and the most essential features.

Internet of Things (IoT), Physical and Machine Learning Products:
These products implement usage of these technologies. The goal here is not just correct implementation of the technologies, but also to ensure that all the product's features cooperate,

enhance one another, and prevent the issues associated with siloed development. Example: Smart Home Security Systems

Marketplace Platforms:
By building the product on the knowledge of the user accumulated over time, the goal is to reduce transaction costs. Instead of technical expertise, here, abilities like an awareness of supply-demand and the psychology of clients are helpful. Example: Amazon Business

How does a PM's career trajectory look like?

The regular process, based on responsibilities handled by a PM looks something like this:



The names/titles might be different for different organizations but the roles and responsibilities at different stages are similar. To help you better understand the tasks at each level, we've referred to article by Dacheng Zhao, a Product Director from MIT who broke down the tasks involved in each step as "levels":

Junior Product Manager:
Given a feature, you can ship that feature

Product Manager:
Given a strategy, you can execute and ship the entire product

Senior Product Manager:
Given a problem, you can form a winning strategy

Lead Product Manager:
Given a space, you can find the right problem(s) to solve in that space

Product Director:
Given a person, you can reliably help them up-level (e.g. Jr PM => PM, PM=>Sr. PM)

Head of Product:
Given a team, you can create the environment and breathing room such that they can progress in hierarchy

Chief Product Officer:
People (PM, Eng., UX, etc) disproportionately want to work with you and on your team.
Now, one cannot define a certain period for moving through the stages, based on the time taken by one to achieve excellence in the roles mentioned above, one can hope to move forward.

Chapter 2: Preparing for the PM role

Do I need to have some technical /coding background?

It is needless to say that Product Management roles are one of the most sought after at MBA colleges. However, with the current curriculum, it is difficult to equip oneself with the skills and technical expertise required to crack PM role interviews. So how should one go about preparing for the same?

One of the things PM companies' values is a technical background or a CSE degree. But worry not, you can still land yourself a PM role without either of these two things. Read further to find out how.

1. ***Technical Background*** – Assuming you have a software engineer profile or have a CSE degree, the interviewers will not go easy on you. Your background will help you land interviews at many PM companies during the process. It is advised that you brush up your fundamentals. You will get detailed information about the same later in this book. So, along with the PM cases, you can expect technical questions.
2. ***Non-Technical Background*** – While having a technical background does give one an edge, it does not mean that one cannot be a product manager without it. A lot of non-technical background folks have thrived in this field. If you can discipline yourself and hone the required skills over time, you can easily land a PM role.

Preparation Tips – How can I build my PM Profile?

PPTs of PM Companies: Several companies give Pre-Placement Talks before the placements. Attending these PPTs is moderately useful. Sometimes they talk about the new products or technologies the companies are building. For those company interviews, it can become an interview question or at least a plus if you know about the company. In either case, their PPTs will be helpful in just knowing more about the company and help you evaluate your interest for the same.

Case Studies/Design Competitions: Taking part in these competitions can be a good way for the company recruiters to get to know your ideas. Sometimes, winning these company competitions can directly award you a PPI. You will also learn PM concepts, especially the application part while solving cases.

PM cases for landing interviews: Many PM companies may provide you with a case to solve and submit a ppt of the same to get their interview shortlist during placements. Preparing well for these will be helpful, you will also need to learn to make basic wireframes that you will have to include in your ppts. Wireframe is a schematic, blueprint, useful to help you and your programmers and designers think and communicate about the structure of the software or feature you wish to develop. Do look out for these activities to get a place in those companies' interview panels.

Forming groups for interviews & GD prep: Preparing for interviews and GDs is recommended in groups rather than individually. You can also try to form a group with similar people who are also looking to get into PM roles. Preparing in a group has the following advantages:

1. Different group members bring with them diverse ways of solving a problem. Some of these solutions might be more efficient and optimized than your solution.
2. You can get to learn from different experiences especially if someone has a technical background and how they dealt with technical problems in the past. This knowledge can be useful in interviews when you are given a problem to solve that you have personally never encountered but you had talked to someone who had been in a similar situation. You can mould your situation to the interviewer according to what you heard from someone else.
3. Tasks can be divided among the group members and conquered together. Given the limited time and energy, you can divide the workload among the group members. For example, one person can prepare daily technical news tidbits, while the other person can look through company websites and make a summary of all their products. Information can be shared amongst group members once all bases are covered by individuals.
4. You get motivation and encouragement. You will get better with practice and hence it will increase your confidence as well. You can collectively grow with the group members.

A good rule of thumb while opting for PM prep as someone from a **non-technical background** is to form *a **group of 3 people*** who are dedicated to PM preparation. All three friends can get together once a day and discuss different PM concepts, solve cases, check out tech news, prepare for some common questions, etc. Of the three people, 1 will be an interviewer, 1 will be an interviewee and the third person will be a spectator. Keep rotating the people to optimize benefits for everyone. Try solving ~2 cases and going through the news every day

Try to research, critique, and discuss different apps widely used by people. This will help you develop and enhance your product acumen. Have the frameworks written down in a document or your whiteboard or your notebook, whichever you find easily accessible. Frameworks are a good way to ensure structural thought processes while solving a PM case or problem. Over time you will adapt to the PM process and be well prepared for the interviews.

The same method of preparation applies to all backgrounds but is essential for people with a non-technical background. Also develop questionnaire common for PM and try to write down your answers for the same. Try to get it peer reviewed and improve upon those answers. These can be resume based questions, HR questions, technical questions, product design questions, etc. No matter the background, all PM enthusiasts must prepare themselves for case studies, questions regarding their past work experience and most importantly, why PM?

Here is a list of topics you should prepare for:

1. Resume based questions
2. HR questions
3. Guesstimates
4. Product Design/Modification questions
5. Technical questions including latest technology news
6. Technical cases
7. Coding/Algorithm based questions
8. Company specific products

A few resources that will help you get started:

1. Cracking the PM Interview by Gayle: This book will help you structure your preparation process. It covers the basics of most of the topics in the list above.
2. Decode and conquer by Lewis C. Lin: Another great book similar to cracking the PM interview. It helps one learn the basics of most of the topics and covers a variety of example cases
3. Case Interviews Cracked by Kelshikar and Garg: This is a case prep book and contains many example cases and their solutions. It provides a good brief about how to solve cases. The knowledge and skills you learn from this can help you solve other PM type questions such as guesstimates, product design/modification questions, and technical cases. The aim of going through this entire book should be to learn how to structure your thinking in a way that it can be used to solve any problem.
4. TechCrunch and any other technical news websites: Useful for keeping oneself up to date with the latest technology news.
5. Company Websites/PPTs: Useful for finding out company specific product information.
6. The Product Folks YouTube/website: You will find a lot of teardown videos of different products that have been uploaded by them on their website and YouTube channel.

This is not an exhaustive list of resources as you will be able to find a lot more useful resources online.

Please refer to appendix of resources for more information.
Do not skip that while reading this section

Does Preparing for other fields help my preparation for a PM role?

This is a frequent question that gets asked. Preparing for fields like finance for example, will not be helpful. Finance prep is knowledge heavy and the knowledge you would be acquiring will be mutually exclusive to a PM role. Prep for marketing may help a little as you might pick up some

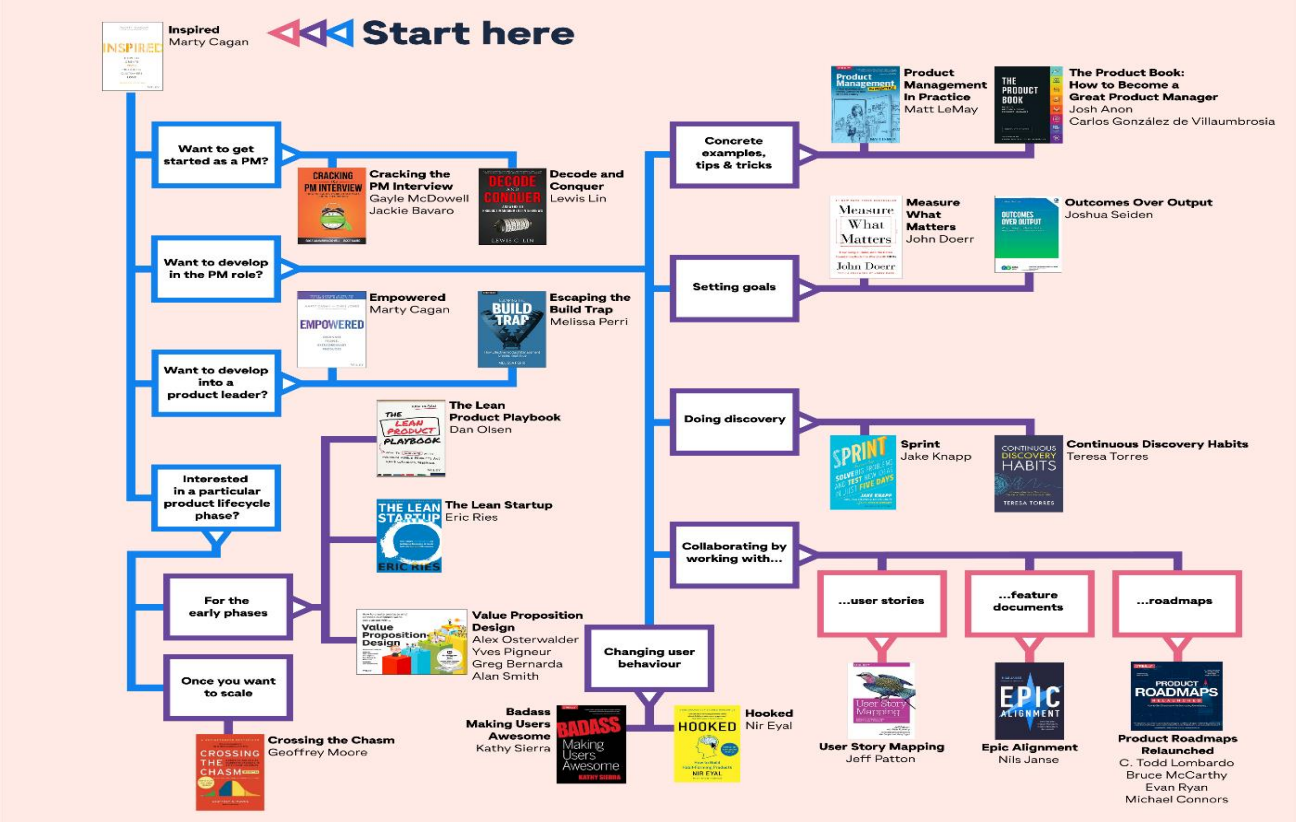
marketing frameworks while solving marketing cases. However, marketing requires different learning skills which are not directly applicable to a PM role.

On the other hand, prep for consulting does help. The reason why prep for consulting can be useful while preparing for PM is that consulting prep teaches you how to structure your thoughts and break down problems to come up with the best solutions despite a lack of information. And that is a useful skill in a PM interview. An interviewer always evaluates how well you can structure the problem given the limited information and how you come up with the best solutions.

Despite consult prep being useful, it is only a subset of the total PM prep and is in no way a compulsory thing. Also, if you do consult prep, be prepared to put in the extra effort to gain knowledge that is specific to a PM role.

If you are into books, you can read the following books for PM topics.

Note: The images are clickable




Getting Started
1. The basics
Design of everyday things
Don Norman
Don't make me think
Steve Krug
User friendly
Cliff Kuang & Robert Fabricant
2. The Mindsetters
Creative confidence
Tom and David Kelly
Steal like an artist
Austin Kleon
The laws of simplicity
John Maeda
3. Design thinking
Design thinking playbook
Michael Lewrick
Jean - Paul Thommen
Larry Leifer
Change by design
Tim Brown
4. Product thinking
Building better products
Laura Klein
The lean product playbook
Dan Olsen
Hooked
Nir Eyal
Specializations
1. Understanding Humans(research)
Talking to humans
Giff Constable
The MOM test
Rob Fitzpatrick
100 Things
Susan Weinschenk
Thinking fast & slow
Daniel Kahneman
2. UI Interaction Design
About Face
Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel
Design for perfect screen
Tal Florentin
Designing interfaces
Jenifer Tidwell, Charles Brewer & Aynne Valencia
Information Architecture
Donna Spenser
Microinteractions
Dan Saffer
Evil by design
Chris Nodder
3. Visual Design
thinking with type
Ellen Lupton
Interaction of color
Josef Albers
Refactoring UI
Adam Wathan & Steve Schoger
4. UX Copy
Microcopy
Kinneret Yifrah
Strategic writing for UX
Torrey Podmajersky
5. Mobile Design
Principles of Mobile App Design
Think with Google
Mobile First
Luke Wroblewski
6. Design System
Atomic Design
Brad Frost
Design system handbook
Designbetter.co
Hack the design system
Capgemini Invent
7.Usability
Rocket surgery made easy
Steve Krug
Mobile Usability
Jakob Nielsen
Process
Sprint
Jake Knapp
Lean UX
Jeff Gothelf
UX Strategy
Jaime Levy
Portfolio
Show your work
Austin Kleon
Job Interviews
Solving Product Design Exercises
Aviam Dashinsky
Leadership
Creativity Inc.
Ed Catmull
Redesigning Leadership
John Maeda
Design Leadership Handbook
Designbetter.co
Business thinking for designers
Ryan Rumsey - Designbetter.co
Curated by
Mohammad Owaiz Shaik
www.owaiz.in

Core PM Topics - Must Prepare Topics

The chapters from 3 to 10 are the must prepare topics to prepare for your summers – Product Design, Feature prioritization, Metrices, Guesstimates, Root Cause Analysis, Dashboards and Behavioral questions. Questions during your summers will revolve around these topics.

First Time Readers – You must skim through this section before starting with the chapters. You do not need to go in depth by clicking on the reference links right now. Revisit this chapter once you are done with major chapters 3 to 9. It is fine if you do not get everything in one reading. Please do not worry! Just try to understand the flow or the sequence of events

Advanced users – Focus more on the adjacent links/references go in detail on each of these steps as this is what is typically expected out of you especially during interviews. So be crystal, clear on this as this is the heart and soul of a PM's job

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Let us first discuss the flow of events that happens typically for a PM:

1. **Problem Scoping**: Clarifying the meaning of the important keywords in the problem and the places where it is applicable – What? Why? Who? How? For instance, say "**We are asked to improve user engagement for a music streaming like Spotify**", some of the clarifying questions could be:
    - What do we mean by engagement here? Is it increasing the user's ability to listen to longer durations of music? Or does it involve interacting more with features on the app?
    - Spotify has both website and app do you want to improve the engagement in website or for the app? If for the app – is there any specific preference for Android vs IOS?
    - Spotify also allows you to listen to podcasts and music. Which one do you want to target?

This is crucial to do in any problem statement. Some ways of going about doing this is mentioned here

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2. **User Segmentation**: Normally we know a very broad set of audience for whom we need to create an App. But we would need to identify what are the different possible segments of users that could be potential users of the application and try to categorize it by defining their characteristics. For the above problem statement, a way of segmenting the users can be found here.

Note: You could also classify users on basis of multiple parameters like age, moods, location, usage frequency or a combination of these to come up with an apt segmentation w.r.t the above problem.

Common methods of user segmentation could be found here. Try to do a MECE for the user segment to ensure we do not miss any potential customer while addressing about the problem

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3. **User Targeting**: Typically, when you have multiple user segments from step 2, on basis of your objective defined in the problem statement, you should do some guesstimate on how large each segment and some secondary research would be to find out whom probably should you target to achieve your objective. In interviews this can be based on your gut feel but with sound logic. For the above problem statement, let's say we target low level Involvement users – so the problem reduces to "***How can Spotify facilitate the ability to listen to curated music for a user who does not want to control music at all times****?"*

4. **Developing customer personas and Identifying Pain Points**: We should take a representative for a set of similar audience and try to find what are the problems they face while using the application. To do this we should put ourselves in their shoes and empathize with the customers. Look at their journey right from getting the app opened till the end:
    - Map it out with how the customers feel during this entire process,
    - List down what portions of the process are painful for the **chosen target segment** at each stage.
    - Is there something that the app is missing that they might be looking for? "Unmet Needs"
    - Is there something that the direct competitors might be offering to them that we are not? Is that something relevant to this chosen target segment?

    Note: A Sample persona for Spotify app would be found here . There could be multiple such personas

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Once the target user segment is chosen, **do not deviate** from that selection while thinking about pain points. This is a very common trap that people fall into, the thought process must be coherent all the way through.

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5. **Prioritizing the User Pain Points:** The above list collated in step 4 could be very huge for each customer persona, and it is impossible to address all of them at the same time.

So, we need to pick. For deciding on how to pick, we can look at the probable severity of the issues and how poorly it impacts customer experience. Based on the above you combination of attributes, you can classify the problems into High, Moderate, and low based on how significantly this problem impacts the user journey. Post which you can choose to address 1-3 most significant problems you feel this user persona would face.

A step by step process on how to go about prioritization of customer pain points can be found here

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6. **Listing down all solutions:** For only the prioritized user pain points in step 5**,** think about various ways to solve the problems in the application. These ways are called features in product management. Do not worry if they are technically feasible or not but should be something that a PM at his level could do. Typical features for the above problem statement regarding Spotify can be found here
7. **Feature Prioritization**: Among the features listed in 7, now is the time to decide on how to prioritize the features based on some frameworks which are captured in Chapter 4. Normally in actual process this is done by talking to the engineering team, doing market study and competitive benchmarking to come up with valid data points to target a particular feature. Discard those solutions which cannot be achieved technically and choose the top 2 features you want. ***Say we prioritized the development of live room feature in spotify***

Ensure the features that you propose is something that the target user segment defined earlier will be able to perform and benefit from. **Your solution cannot go against the pain point and target segments chosen earlier**
DO NOT PROPOSE marketing solutions, partnership/collaborations, changing the strategy/vision for the product unless the question specifically ask for it.

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8. **Wireframing & Task Flow**: Coming up with how the feature would look like and operate for a user. For Spotify the task flow can be found here and the wireframes are:
    - Creating a live room
    - Live Room Home Page
    - Interaction in Sub-Community

Once you have covered chapter 4, you can come and revisit this page to see some when to use which framework on a case to case basis

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9. **Metrices**: Once you have defined the top 2 features decide on how to measure those features, coming up with measures for the same using some of the frameworks defined In Chapter 5 of this casebook.

10. **Product Implementation**: Once the features are finalized, come up with how you would implement those features, some sort of timelines and making sure you keep a tab on the commitments. Normally this is skipped in the interview but is a requirement in day-to-day job.

Note: The example of Spotify is taken for illustration purpose, and it currently has this live room feature in its application, but this is how they went about developing it.

The above steps form the basis of how you go about reaching the solution from a problem statement in a typical PM situation.

We will now be revising most of the concepts covered in these 10 steps in Chapters 3 to 9 to get some more flavor on how to solve typical such problems in the world of PM.

Chapter 3: Product Design

The questions need to be answered in a very structured manner, even though they might seem like open-ended questions. Understanding user needs is most important here. This type of question is one of the most asked questions as they evaluate your creativity, ability to empathize with users, ability to connect the company's goals with user needs, and prioritization skills, all of which are core to being a good product manager. Mainly of 2 types – Improve a product or create a new product from scratch. The products can be physical or digital

Examples of Product design/ improvement questions

1. *Design an intelligent table for restaurants.*
2. *How would you improve WhatsApp?*
3. *You are a PM at Spotify; how would you increase user engagement?*
4. *Design a smart shoe.*
5. *Design an application for trek enthusiasts.*

Common mistakes candidates make when asked product design questions

1. **Jumping to features directly** - Listing features and giving solutions without setting up an appropriate context shows that you haven't thought about the answer holistically. The listener doesn't know who the intended customer is or what problem you are trying to solve. You won't be able to justify your solution among other possible solutions.
2. **Unstructured answer / Rambling** - If you haven't organized your thoughts before speaking, you will end up improvising as you go, which is painful for the listener. The solution won't be impressive, and you will invite many cross-questions.
3. **Not asking questions** – In case of vague problem statements or questions about a product you are not familiar with, it is better to get more clarification and understand the expectations through clarifying questions.

The following framework will help you answer any type of design question. The sequential structure of the framework will enable you to move through essential questions to understand what to design fully. It will act as a checklist to form an exhaustive and organized response to design questions.

CIRCLES Method

C – comprehend the situation

I – identify the Customer

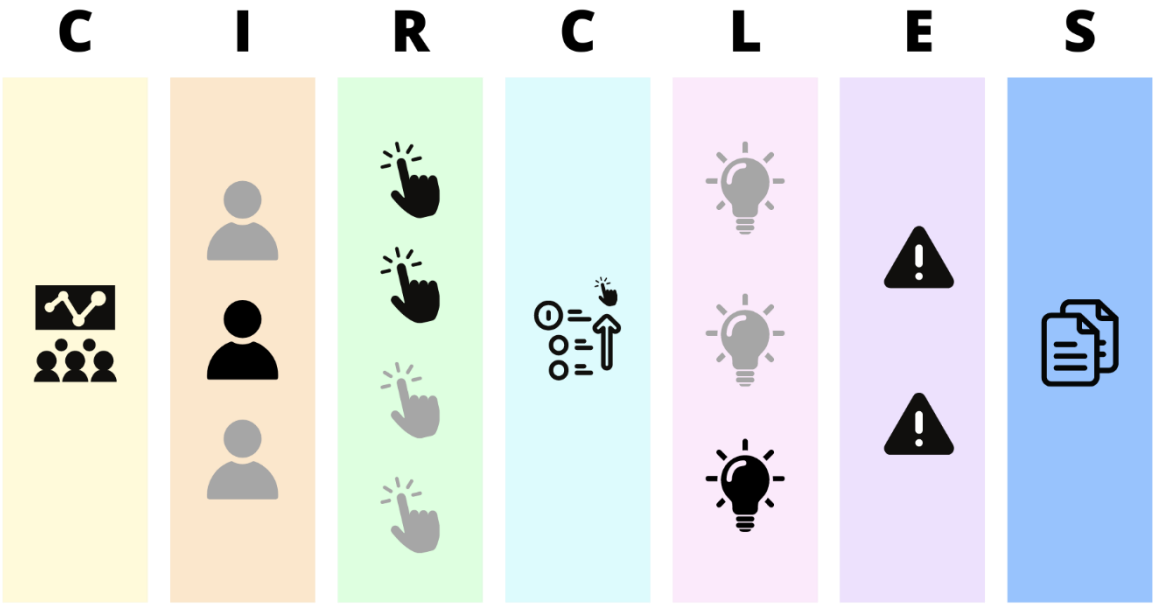
R – report the customer's needs

C – cut through prioritization

L – list the solution

E – evaluate trade-offs

S – summarize your recommendations



1. **Comprehend the situation**

   A product manager should seek to gather as much information as possible about a design question to understand the context better. You can clarify the goal (e.g., increase revenue, market share, or engagement), understand the constraints, if there are any, and understand the context (e.g., what is it? who is it for? Why do they need it?).

2. **Identify the Customer**

   You don't want to build a product that does everything for everyone; hence focusing on one customer target segment is essential while answering. You are suggested to list potential consumer personas for the product and explain their characteristics briefly. A comprehensive approach and listing of interesting personas will fetch you the brownie point. Some examples of customer personas are:

   - College students
   - Professional athlete
   - Trek enthusiast
   - Food lover
   - Small business owner

   Now you need to choose one persona to focus on; one way to prioritize the user persona can be based on the total addressable market and the extent of unmet needs. Explain whom you are talking about briefly using the following characteristics.

Persona name	Behavior	Needs and Goals	Pain Points
Kat, the traveling reader	• Goes on vacations with books • Travels four times a year • Carries four books per trip	• Discover new books • Discuss books with others • Write a book one day	• It is difficult to find good books to read • It is difficult to get into the habit of writing

3. Report the customer's needs

The third step of the CIRCLES method is reporting the customer's needs. You can define customer needs in a use case or user requirement format. The following template can be used

As a <role>, I want <goal/desire> so that <benefit>.

e.g., in the case of user persona Kat, the user story can be written as follows

a. As a traveling reader, I want to get recommendations so that I read books that are either well-written or are good examples of my favorite genres.
b. As a traveling reader, I want to write 500 words daily to publish my memoir.

4. Cut through prioritization

With the given time constraint in the interview scenario, you cannot design a solution for every pain point you discovered in the previous step. Hence, among all the customer needs to be listed, you must prioritize one or two customer needs. One way to prioritize can be based on how acute the problem is (depth) and what is the frequency (breadth). Another method can be found on customer value generated and effort; you choose a pain point that will create maximum customer value if solved with low or medium effort.

The prioritization step mimics the real-world development process. You'll have a significant backlog of use cases, but you're limited by time, money, and labor. Which one do you do first?

5. List solutions

This is undoubtedly the most fun part of the answer; this is your opportunity to show off your passion for product management. A customer pain point can be addressed in various ways. For example, for book recommendations, there can be a recommendation algorithm or a feature that lets your friends recommend books to you on the app. Also, if you have chosen more than one user pain point, you will list solutions for each one. These solutions must be features you can include in the product that will solve your customer's needs. Thus, you will have a short list of solutions (not more than 4). Remember that most employers

are evaluating your creativity and product vision. They're looking for product managers who can see future trends in technology and customer behavior. They expect those product managers to plot and execute a plan that exploits that trend for the company's benefit.

Once you've listed all the solutions, you can prioritize the solutions based on popular frameworks such as RICE, Value vs. Effort, Kano model, MoSCoW Method, etc. These frameworks will be explained in the latter part of the book.

6. **Evaluate trade-offs**

   The second last step of the framework is to evaluate the trade-offs of your solutions. You can mention the pros and cons of the solution. For example, suppose you have mentioned a rating system for an accommodation app so that users can look at the reviews from previous guests and know what to expect. In that case, it may also lead to issues where dishonest reviews might hurt the owner's business. By describing the trade-offs, you come across as thoughtful, analytical, and objective. You'll also protect yourself from being defensive. If you've taken the initiative to critique your own solutions, the interviewer has fewer things to criticize.

7. **Summarize your recommendations**

   In this optional step, you can take 20-30 seconds to summarize your answer using the following three steps

   1. Which feature do you recommend?
   2. Recap on what it is and why it is beneficial for the users and/or the company
   3. Why do you prefer this solution over others?

Understanding the CIRCLES method and more importantly practising multiple questions on it, is the key to cracking the PM interviews. Watch this video to get more clarity.

Practice Question 1 – Product Improvement

Interviewer – How would you improve engagement for Headspace?

Candidate – Here's how I would like to go about this problem statement

- Headspace's Mission
- Users and their use cases
- Pain points
- Solutions and prioritization

First, I'd make sure that I understand the company's mission; then, I'd explore the user and use cases; and lastly, I'd prioritize the use cases and focus on brainstorming solutions.

Interviewer – Sounds good.

Candidate – Headspace's mission is to improve the health and happiness of the world. People use Headspace to meditate, and engagement for Headspace can be measured as no. of a meditation session or an average number of minutes meditated.

User segment can be divided into three groups – Skeptics, Rookies, and Die-hards

Skeptics – people who may be tried Headspace a few times but didn't resonate with the idea.

Rookies – Tried Headspace a few times, saw some value but could not form it into the habit and getting into the grove of meditating regularly.

Die-hards – Those people who saw the direct impact of meditation on their life have already formed a habit. Perhaps they are the advocates of the app and meditation.

Does this sound good?

Interviewer – Yes, that makes sense.

Candidate – I would now like to take a minute to prioritize the user segment we will be focusing on. Two criteria come to my mind.

1. TAM – total addressable market
2. Unmet needs

From a TAM perspective, I am thinking about people who are actually using Headspace as we are targeting engagement here. In this context, rookies are the most significant segment with maximum potential to increase engagement. Die-hards are already seeing the benefit, and incremental engagement will be less. Skeptics have higher friction and may not open the app very often.

In case of unmet needs, rookies can still not form a habit of meditation and get maximum value from the app.

Moving to paint points of rookies, I am thinking about the user journey



Paint points

1. Forgetting to meditate
2. Concentrating during meditation
3. Understanding the direct impact of meditation

Premeditation – most significant pain point, just getting into the phase to remember to meditate. Usually, people are so busy and get caught up in day-to-day activities that forming a meditation habit is difficult.

During meditation – for those who remember to meditate and make an effort to do it while you meditate, perhaps the most significant pain point is that it's difficult to concentrate.

Post meditation – If you are a rookie, it is somewhat difficult to see the direct impact of meditation on day-to-day life as it's a slow process, and one must be consistent.

Now, I would like to prioritize the pain points based on two factors, how acute and how frequent the problem is.

I think the first pain point is most frequent and will prevent a rookie from becoming a die-hard.

I think potential solutions to address this pain point can be **Support Group, Score Board/Leaderboard with Friends, Loyalty Program**, which is explained in detailed as follows

1. **Support Group** – With a support group, you can build a rapport with someone facing similar difficulties, and it's harder to disappoint someone than to disappoint yourself. Create a support group where you can have live meetings, reflection meetings, and weekly cadence. Meditating with a group will help to develop a habit. Also, in the support group, you can find other practitioners of varying levels; they can help each other clear up doubts and confusion regarding the best practices.

2. **Scoreboard/leaderboard with friends** – In leaderboard functionality, users can simply see who among their buddies has used the app on that day or how many times a week. If they haven't used it, users can just send them a nudge and remind them to meditate. The app can provide badges such as longest streak and most levels mastered that will motivate users to maintain consistency and help them track their progress in the app.

3. **Loyalty Program** – Headspace can develop a loyalty program where users can access special sessions with expert practitioners, in-person seminars, and meet-up events to build a strong community. The app can also send merchandise such as stickers to users enrolled in a loyalty program that will remind them to meditate.

I would like to pause here for a moment. If you have any questions or suggestions for me, please let me know

Interviewer – I agree with some of the solutions you have come up with. Can you tell me which one should be implemented first?

Candidate - Yes, now I would like to move on to prioritization. I will evaluate the solutions regarding the effort required and the impact created. The feature that makes the most impact on the required effort should be implemented first.

The scoreboard/leaderboard looks like a solution with maximum impact for a given effort. A support group will require us to curate the mechanism of group creation and facilitation of interactions and work in the long term once we see strong network effects. The loyalty program may have limited reach in the initial phases, and it will be challenging to increase engagement for the majority of users through such as program.

Practice Question 2 – New Product Development

Interviewer: Design a smart table for a restaurant

Candidate: First, I would like to consider the restaurant space in general, the types of restaurants, and what we can offer that is differentiated. These can be different types of restaurants

1. Fast Food Restaurants
2. Casuals cafes, clubs, and restaurants
3. Mom and pop restaurants
4. High-end restaurants (cuisine specialties)

Thinking of smart tables, some fundamental functionalities can be

- Order food
- Call you waiter
- Track the order
- Entertainment

I would like to narrow down to a good intersection of type of restaurant and touchscreen or smart table

In fast food restaurants, people don't want to spend too much time; Also, mom-and-pop restaurants are generally family owned and operate on small margins; when it comes to high-end restaurants, the atmosphere is sophisticated, dimly lit, and quiet. They want to provide a quiet and intimate dining experience. Not ideal for the potential sounds and lights of a touchscreen table. Hence all these types of restaurants won't be the first targets

Casual cafes, clubs, and restaurants are perfect opportunity; these places can have a capital expenditure budget and aims to provide casual, fun time with family and friends.

The product's mission can be - To design a touch screen table for casual restaurants and clubs that want to provide a fun and friendly dining experience to their customers.

Moving on to use cases and types of users. Following are some of the potential user types

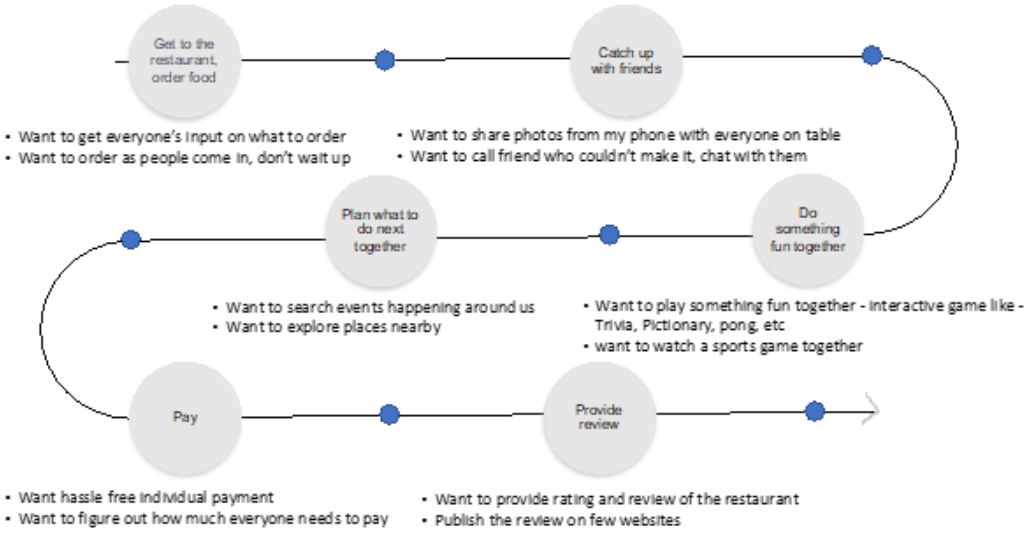
1. Couples dining on the dates – Want to focus on each other's company, hence no significant need for entertainment.
2. Friends get together - unmet needs, want to socialize and have fun in novel ways. A large number of diners.
3. Families eating out - unmet needs, primarily to entertain kids. However, some existing solutions like smartphones do that.
4. People celebrating special occasions like birthdays, and anniversaries - unmet needs, can help them to make this event special.

I will pick ‘friends get together’ as the product's mission is to provide customers with a fun and friendly dining experience.

Can I go ahead with this user segment?
Interviewer: Yes, please go ahead!

Candidate:

User journey and their associated needs can be described as-



Some of the solutions to address these needs can be

1. Transforming the dining experience with the help of games and interactive activities
2. Sharing of digital content via a common network similar to screen share
3. Next generation of food ordering – Digital menus, recommendations, food picker, tracking order, customizing orders
4. Digital payment and review integration

I will prioritize the solution based on the impact on the user's goal and the complexity of development

	Value	**Effort**
Games and interactive activities	High	High
Sharing of digital content	Medium	Medium
Next generation of food ordering	Medium	Medium
Digital payment and review integration	Low	Low

Based on the prioritization, I want to focus on 'Transforming dining experience through games and interactive activities' because it fits into the goals of our target segment and brings a new way of thinking about the restaurant experience.

Solution Details

1. The table has an interactive menu where you can pick a game to play. Categories like card games, board games, trivial, drawing games, etc
2. You can invite other friends who are not at the table with you; they can join you virtually
3. Once the game is selected, the screen's layout changes to suit the game.
4. Can integrate with an app store to download new games

Lastly, some of the Trade-offs of this solution will be

1. People spend more time at the table for activities other than eating
2. The layout should be designed to make sure there is enough space for food, plates, glasses
3. Sound from these games can disturb other people at the restaurant

Interviewer: Very well thought through, that will be all. Thank you.

Solved Examples

1. Improve Whatsapp
2. Improve Instagram Stories
3. Improve Air Travel
4. Improve a Fin Tech Product
5. Design a Parenting Product
6. Design a Photo App for the Blind
7. Design a Fitness App
8. Design a Product for Rush Hour Drivers

Chapter 4: Feature Prioritization

Product managers must decide where the product moves next. But one cannot do this in isolation, many voices of various stakeholders weigh in and customer feedback guides what to build next. Without a prioritization framework, the product manager would be relying too much on subjectivity and fail to reach stakeholder consensus. Effective prioritization provides transparency and builds confidence in the planning process.

The following are a few commonly used frameworks:

A. RICE

This scoring methodology measures every feature or initiative based on four factors: reach, impact, confidence, and effort.

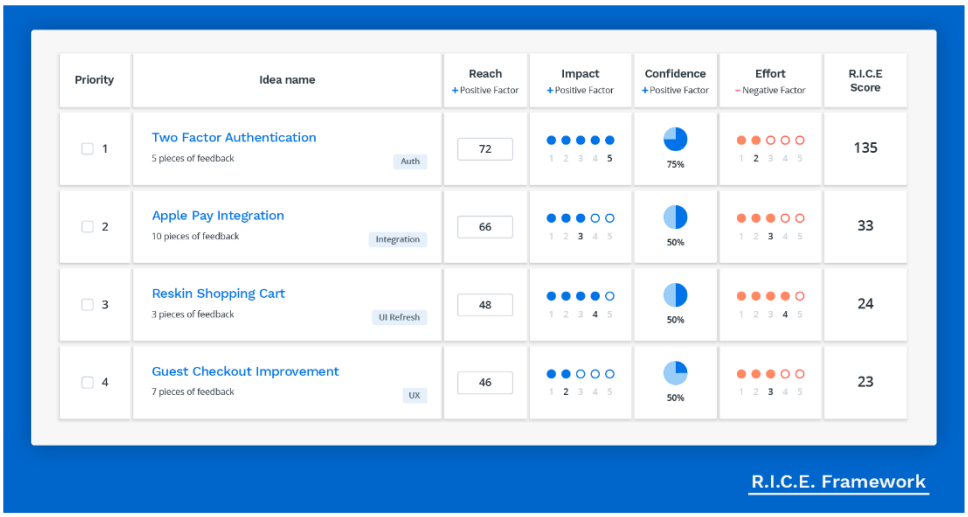
REACH	IMPACT	CONFIDENCE	EFFORT
How many people will this feature affect within a given time period? **Example:** customers per quarter, transactions per month	How much will this impact individual users? Use a multiple choice scale: **3** = massive impact **2** = high impact **1** = medium impact **0.5** = low impact **0.25** = minimal impact **Example:** How much will this feature affect conversion rates?	How confident are we about the impact and reach scores? How much data do we have to back up those estimates? Use a % score where: **100%** = high confidence **80%** = medium confidence **50%** = low confidence	How much of a time investment will this initiative require from product, design and development? Measure as persons per month (how much work one team member can do in a month).

Individual scores output an overall score through RICE formula. It gives a standardized number to compare various features or initiatives to compare.

$$\frac{\text{REACH} \times \text{IMPACT} \times \text{CONFIDENCE}}{\text{EFFORT}} = \text{RICE SCORE}$$

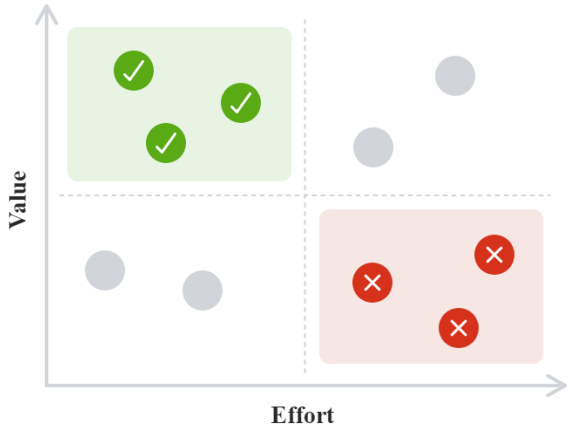
Do not confuse between Reach and Impact – Reach Talks about how many people will this impact? and Impact talks about how much will this impact each person

Example:



B. Value vs. Effort

This is simplified version of RICE which is also known as the value and complexity framework and gives an objective way to determine which initiatives or features to prioritize. Each action is scored according to the value it would bring to the product and level of implementation difficulty. It is built on a 2x2 prioritization matrix,



While estimating scores for value and effort, product team can think of these guidelines:

How important is it?	How difficult is it to build?
• **Value** • Potential revenue • **Benefit** • To current customers • To potential customers • **Impact** • On the business goals • On the strategic goals	• **Cost** • **Effort** • Development effort • Operational effort • Implementation effort • **Risk** • **Complexity**

C. Kano Model

Kano model believes that Customer Satisfaction depends on feature's level of functionality. It has two dimensions:

Satisfaction, alternatively Delight/Excitement ranging from Total Satisfaction (Delighted or Excited) to Total Dissatisfaction (Frustrated or Disgusted).

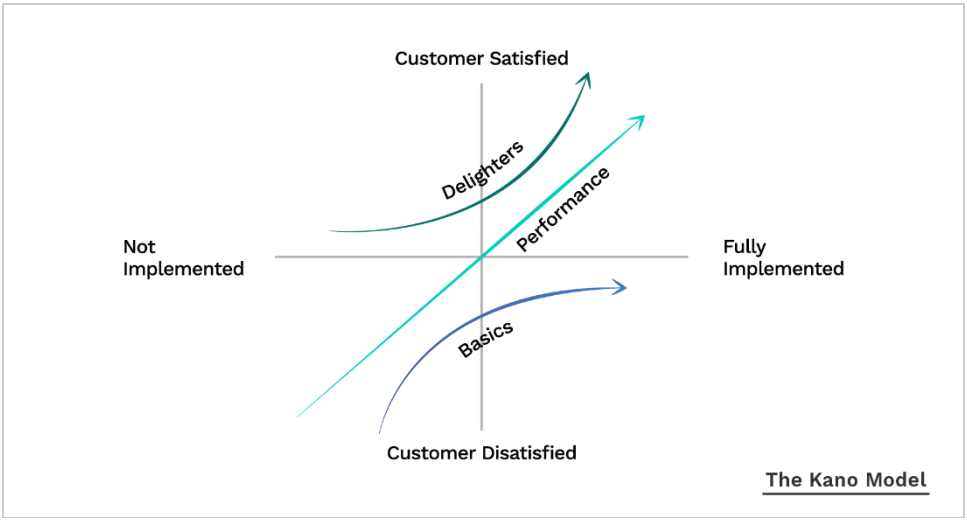
Functionality, alternatively Achievement/Sophistication depicting how well a given feature is built. It ranges from Didn't Do It at All (None or Done Poorly) to Did It Very Well.



It classifies features into four categories:

1. **Expected (Must-have),** some features are expected as basic functionality.
2. **Normal (or Performance enhancers),** the more of such features added, the more satisfied customers are.
3. **Exciting (or Attractive/Delighters),** provides delightful customer experience but does not add value functionality-wise.

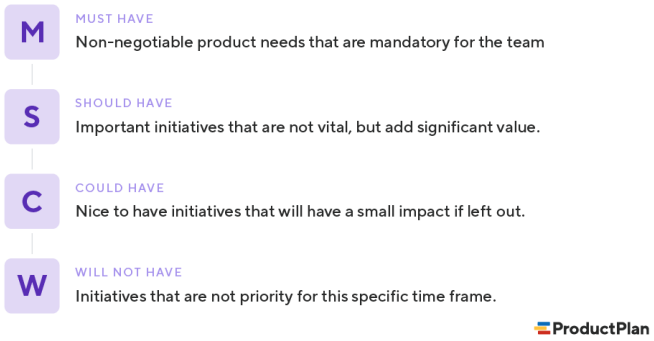
4. **Indifferent,** the presence or absence of some features doesn't impact the customer value.



D. MoSCoW

The MoSCoW method classifies features into four priority buckets. MoSCoW stands for **M**ust-Have, **S**hould-Have, **C**ould-Have, and **W**on't-Have features.

1. **Must-Have:** These are the features essential for a product to be functional. These are non-negotiable and essential. Without fulfillment of all of these, a product cannot be launched, thus it is the most time-sensitive of all the buckets.
    a. Example: User login feature
2. **Should-Have:** These requirements are important with respect to delivery but are not time sensitive for launch.
    a. Example: Reset password feature
3. **Could-Have:** This feature is not essential or important to deliver within a timeframe for launch. Example: Single sign on
4. **Won't-Have:** These are the least critical features, tasks, or requirements and would be considered for future iterations or releases.





E. ICE

Similar to RICE model, ICE model scoring model based on Impact, Confidence and Ease of development. It is also referred to as a minimum viable prioritization framework.

Impact – Impact of an initiative
Confidence – Confident of initiative's ability to prove hypothesis and deliver the desired results
Ease – Ease of implementation

These factors are generally scored from 1 to 10, and the total is averaged to get the ICE score for a particular feature. An example of ICE prioritization can be:

Project idea	Impact	Confidence	Ease	Ice Score
Billing System	7	1	5	35
Community tab	7	2	8	112
Update receipt design	5	5	3	75

F. Other Frameworks

There are multiple other frameworks like – Story mapping, Opportunity scoring, The product tree, Cost of delay Affinity grouping, Buy-a-feature, weighted shortest job first, which help product managers to smoothly prioritize between competing requirements and plan product execution strategy. In case you are interested, you can look at these frameworks on the net.

Solved Examples

1. Explanation Videos can be found here for more prioritization frameworks.
2. A sample interview for product prioritization can be accessed here.

Chapter 5: Metrices

HEART Framework

Use case: This is a framework to evaluate a specific aspect of user experience. It is generally used as a framework to determine areas to prioritize while developing new features or products
Origin: Google UX team

Direction for use: HEART is an acronym that denotes the 5 key aspects for a given features for every category we need to define Goals, Signals & Matrices.

	Goals	Signals	Metrics
Happiness			
Engagement			
Adoption			
Retention			
Task Success			

Goals are broad objectives. For Happiness, the Goal could be to "increase user satisfaction". Signals are an indication of progress. For example, "users spending more time per session in our software". Metrics are Quantifiable data points that can indicate success of failure. For example, "Churn rate" is a useful metric for Retention.

Using these metrics can help you objectively prioritize activities for your project. It is however not necessary to always use all the parts of the Heart framework. In some cases, just four or three categories are sufficient.

Check out the Video Link for better understanding: https://www.youtube.com/watch?v=ulQ3NKPBCt0 (10 Min)

PIRATE Framework (AARRR Framework)

Use case: To study user behavior using matrices relevant for product led growth business.

Origin: Dave McClure, a Silicon Valley investor and founder of 500 startups. He developed these matrices to drive focus to important only those matrices that are important to the growth of the company and to prevent distraction from superficial matrices like social matrices

Direction to Use: The AARRR framework is yet another acronym. It's also known as pirate framework due to how the sound "AARRR" is frequently associated with pirates in popular movies and shows.

The acronym stands for the following:

Acquisition (or awareness) – How are people discovering our product or company?

Activation – Are these people taking the actions we want them to?

Retention – Are our activated users continuing to engage with the product?

Referral – Do users like the product enough to tell others about it?

Revenue – Are our personas willing to pay for this product?)

The objective of using this framework is to focus our attention on the one matric that matters. At every stage of the customer's journey, we will need to track one specific metric.

Given below are some of the many matrices one could consider in different phases:

Acquisition	Activation	Retention	Referal	Revenue
•Number of App visits •Stays over a minute •Click thorugh rate from instagram ads •Conversion form Landing page to app store	•Newletter subscription •Subscribtion for free trial •Succesful onboarding	•Visits app more than 3 times in a month •Opens email and click thorugh •Churn rate •Dail Active Users (DAU) •Monthly Active Users (MAU)	•Sends 1+ user to site •Refers 1+ user who activate •Net Promoter Score NPS •Number of Invites sent •Number of app installs per invite	•Generates minimum revenue •Average order value per user •Customer Lifetime Value

Check out the Web Link for better understanding:

https://medium.com/@ginoarendsz/an-introduction-to-the-aarrr-framework-b8570d6ae0d2 (10-min read)

North Star Metric

Use case: The North Star Metric is a metric a company uses as a focus for their growth. The objective is to guide our decision making in line with the long-term objectives of the company. The general idea is that in the long term if a company brings more value to its customers growth is assured.

Well-known North Star Metric examples are:

- Spotify = 'Time spent listening '
- Airbnb = 'Number of nights booked '
- Facebook = 'Monthly active users '

Check out the Web Link for better understanding:
https://growwithward.com/north-star-metric/ (10-min read)

Common Performance Metrices

Name	Definition	Calculation
Trial Customer Conversion Rate	To measure the rate at which the number of free users convert to paid user	$\frac{\#\ of\ paying\ customers}{\#\ of\ trial\ users}$
Retention Rate	% of customers that continue the platform from one period to next	$\frac{\#\ users\ in\ period\ 2}{\#\ users\ in\ period\ 1}$
Churn Rate	It's the inverse of the Retention rate	$\frac{\#\ users\ in\ period\ 1}{\#\ users\ in\ period\ 2}$
Daily Active Users (DAU)	average number of users in a day	Depends on the product
Weekly Active Users (MAU)	average number of users in a week	Depends on the product
Monthly Active Users (MAU)	average number of users in a month	Depends on the product
Stickiness	Measure of how often users engage with your product (Several times a day or few times a month)	$\frac{DAU}{MAU}$
Customer Acquisition Cost (CAC)	This is the cost incurred for acquiring 1 unit customer	$\frac{Marketing\ spend\ in\ a\ period}{Total\ No\ of\ customers\ generated\ in\ the\ period}$
Customer Lifetime Value (CLV)	The total revenue that can be earned from an average user	$\frac{Revenue\ per\ User\ in\ a\ period}{Churn\ rate\ for\ that\ period}$
Net Promoter Score NPS)	The percentage of users likely to recommend your product (promoters– users not likely to recommend your product (detractors)	$\frac{\%\ of\ promoters}{\%\ of\ detractors}$
Session Duration	total time users spend in your product, divide it by total users, and take the mean value.	
Bounce Rate	the percentage of users who visited only one page of a website or app and left	
Number of users per session	This metric helps understand key user behavior: how often users come back and use the site.	
Number of users action per session	It displays which actions a user made and which feature(s) they used while using the app. This metric is used to understand the popularity of a certain feature since it was introduced and compared to a particular period of time. Also, you can compare these metrics of churned and retained customers and get an idea of what makes the users interested in your product.	

Practice Question 1

Evaluate WhatsApp's in app Video Playing Feature?

Step 1: Describe the feature and its underlying goals

WhatsApp's in app video playing feature allows users to watch a particular YouTube video from within the chat window itself. The primary goal of this feature is to improve customer engagement on the platform

Step 2: Explore the customer Journey (specific to the use of the feature). A sample customer journey is given below:

Step 3: Quantify the action taken by the users

Step 4: Summarize your recommendations

Here the success of the feature depends on how well we can improve the engagement of customers with WhatsApp. Identify which of these are the most important metrics in this context and therefore prioritize them accordingly. In this example as our objective is to improve engagement, the average time spent per user & number of users using the new feature can be our most important metric based on the goal we have set.

Practice Question 2

As a product manager at a leading E-commerce company, what are the metrics you will observe?

Step 1: Describe the products and its goals: An ecommerce company can operate with just a website or using a mobile application. Identify key features of the business and its goals. For a leading e-commerce organization, the key goals are either revenue or profit margin (or both)

Step 2: Explore the customer Journey (End to End customer experience)

A sample *customer journey* for a website is as follows:

1. Customer uses google search to identify a particular product
2. The customer visits the website
3. Customer views the product page
4. Customer completes user registration
5. Customer adds products to cart
6. Customer Pays for their products
7. Customer reaches out in case of issues

Step 3: Identify relevant matrices using any relevant framework based using customer journey

Acquisition:

- Google search organic impressions
- Click Through Rate

- Click Through rate on paid ads

Activation:

- Bounce Rate
- Pages/ session
- Time per session

Retention:

- Number of new users in a given period
- DAU- Daily Average users
- MAU- Monthly Average users
- Churn Rate
- Stickiness

Referral:

- App referrals/ user
- User feedback

Revenue:

- ROI
- Customer Lifetime Value
- Cost of Acquisition
- Number of Items per Cart
- Value of Items per cart
- Conversion from Add to cart to purchase
- Cancelled purchases
- Product Return Rate
- Refund Rate

Step 4: Summarize your recommendations

Solved Examples

1. Metrices for Facebook Marketplace
2. Metrices for Social Media

Chapter 6: Guesstimates

You might be asked to estimate a business-related number as part of a "sizing" or "estimation" question during product management interviews. These questions can be termed as Guesstimates, which are questions that ask for a specific number as an answer but require you to make educated guesses and apply logic to arrive at that number. These questions evaluate your quantitative skills as well as your ability to effectively deconstruct a challenging situation to arrive at a viable solution. This type of question requires some practice, but once you get the hang of it, it's straightforward.
Before we start, a couple of things to remember about these questions:

1. You are not expected to have previously considered this metric.
2. They are questions to which your interviewer is unlikely to have a satisfactory response.
3. On the job, estimates are used (though not every day!). However, they are beneficial exercises for demonstrating your logic in problem solving with very little or no information available to make the correct decision.

- *"When would be the right time release this product?"*
- *"Should the marketing campaign put emphasis on these features?"*
- *"What theme should my product follow?"*

These are some of the problems that you might need to solve without proper data and information. And hence the ability to come toa solution with proper reasoning and logic pays up.

Practice Question 1

Q. How Much Money Is Spent on Gas In The Us Every Year?

Step 1: Breakdown the question and seek clarification to questions
The first step to solving this problem is to take the question, break it down word-by-word, and focus only on the words that are important. How much money is **spent** on **gas** in the **US** every **year**. The 4 bolded words are important. They mean that you need to come up with a yearly demand/consumption of gas in US which can be later converted to amount using appropriate conversion metrics.
Here are some crucial clarification questions to ask:

- Should we solely look at gas use from road transportation? Or does it also include things like house heating and plane travel?
- Should we limit ourselves to personal automobiles? Or should we include usage from business cars as well?

These questions can assist you figure out what your interviewer is searching for. They're also a great way to buy yourself some time. Your subconscious mind may start working on step two of the approach while your conscious mind is asking these questions.

Step 2: Map out your calculations

Here is a detailed process for calculating that number:

1. Find out how many personal vehicles there are in the US.
2. Calculate the normal utilization of a personal vehicle.
3. Add up your consumption in dollars and then in gallons.

If you can't think of a beginning point, an alternate way is to start with the number you wish to compute and create an issue tree from top to bottom, as shown below. Furthermore, the advantage of this technique is that it forces you to be mutually exclusive and collectively exhaustive (MECE).

Both strategies can be considered acceptable. However, before beginning to compute figures in either scenario, you should VALIDATE your strategy with your interviewer by outlining the various calculating procedures you plan to use (step three). This is crucial because it provides the interviewer with the opportunity to change your line of action if they have another idea in mind.
It is exceedingly hazardous to skip this validation stage since your interviewer can become aware too late in the exercise that you are tackling the problem differently than they desired. In our experience, it's really difficult to bounce back from that circumstance.

Step 3: Form the equation and calculate

Once you've chosen a course of action, it's time to begin your calculations. You'll need a lot of data points to get at your final estimate (for example, the amount of miles travelled by an average US motorist each year). Most of the time, your interviewer will ask you to create your own hypotheses in order to determine the final number. They could, however, provide you access to data in certain circumstances.

It is crucial to make assumptions using simple math. For instance, utilize 350 days each year rather than 365. You should figure that a gallon of gasoline typically costs $3.00, not $2.50. Etc. Making computations more straightforward and error-free by rounding numbers.
Additionally, when making computations, talk loudly so that the interviewer may comprehend your thought process. Your interviewer is more interested in what you are thinking than in the outcome.

Step 4: Sense-check your answer

At the conclusion of stage three, most applicants quit speaking. They glance up and anticipate hearing from their interviewer whether they arrived at the correct conclusion or not. This is an error. Before saying they are finished with the interview, the top applicants double-check their work and look for their own errors.
Mental calculating mistakes are common. If the interviewer finds an error in your figures, it will undoubtedly work against you. You still have a chance, though, if you recognize your own error.
Let's take an example to illustrate the strategy you should employ when responding to inquiries about product estimation.

Here are a few things to consider:

- Although the data may appear imprecise, keep in mind that you won't have access to the internet at that time, therefore all the figures and equations you use must be ones you can recall off the top of your head.
- Tell the interviewer in detail every assumption you are making during the computation process. It is not a test of accuracy to solve guesstimates.
- Your method for answering the question—rather than the answer itself—is what is important in this situation. Therefore, be sure to fully explain your thought process to the interviewer even if the response isn't quite accurate.

You might be thinking this is something I have already looked while preparing for consulting interview. So, what is the difference over here. The questions are more focused towards the guesstimate of digital products just like the next one. However, the approach and the way to solve is quite similar. In case, you want more solved examples apart from this two to develop your approach, you can look the material of consulting for guesstimates section

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Practice Question 2

Q. How much revenue does Youtube make per day?

STEP 1

Here are a few clarifying queries regarding this estimating problem that immediately come to mind:

1. Do you want us to estimate YouTube's whole revenue? Or is the only source, which is unquestionably advertising?
2. On YouTube, do advertisers get paid per 1,000 views or each click? Do we care more about YouTube's global or simply US earnings?

Let's say we just want to calculate YouTube's advertising revenue ($) in the US. Let's assume for the sake of simplicity that YouTube charges for advertising largely per 1,000 views.

STEP 2

To calculate YouTube's daily revenues in the United States, we may employ a three-step procedure.

1. Determine the number of daily YouTube users in the United States.
2. Determine how many commercial videos they see and how much money they make consequently.

Before you begin your calculations in a live interview, you should describe your technique to your interviewer and ensure that it makes sense to them.

STEP 3

A. Determine the number of daily YouTube users in the United States.

The United States has a population of 300 million people. Assume that the population is evenly distributed throughout all ages and that the average life expectancy is 80 years old. We need to figure out how often each age group views YouTube:

- 0 to 10 years old — does not watch
- 10 to 20 years old — every day
- 20 to 40 years old — every day
- 40 to 60 years old — 75% of the time
- 60 to 80 years old — 50% of the time

The number of users must then be computed for each age group. If you split 300 m by 4, you get 75 m, and if you divide it by 8, you get about 40 m. For age groups of 10 years or less, we split by 8; for age groups of 20 years or more, we divided by 4.
Age group:

- 0 to 10: 0
- 10 to 20 years old — 100% x 300m / 8 = 40m
- 20 to 40 years old: daily — 100% x 300m / 4 = 75m
- 40 and 60: 75% of days — 75% of 75 minutes, or 55 minutes.
- 60 and 80: 50% of days — 50% x 75m, or 40% of days.
- Total length — 40 + 75 + 55 + 40 m.

The number of everyday YouTube users in the US is therefore around 200 million
It's a good idea to back up your assumptions in a real interview by relating them to your own experience. You may say, for instance, "Since my parents watch YouTube on average seventy-five percent of the time, I'm going to assume that the 40 to 60 age group does as well. And given that my grandparents used to use it so regularly, I'll assume 50 percent for individuals between the ages of 60 and 80 watch it."

B. Determine the total number of advertisements watched.

Let's now determine the daily average for YouTube ad views.

1. Viewers' video viewing habits are likely to differ substantially. YouTube users who stream music often watch 20 or more videos every day. Only one video may be seen by other users. Let's assume that you typically view 10 videos every day. 2 billion views per day are possible with 200 million users.
2. At the beginning of each video, YouTube does not show adverts. Let's assume, based on our observations, that only 50% of videos watched start with an advertising. This translates to 1 billion views of commercial videos daily.

C. Determine the associated revenue.

If YouTube charges advertisers $10 for every 1,000 views, the total daily income is $1 billion x $10/1,000 = $10 million. As a result, we project that YouTube only produces roughly $10 million in daily ad income in the US.

STEP 4

As was previously said, it is imperative that you take the time to sense-check your findings. In practice, you should verify both your final outcome at the end of the estimating procedure and your intermediate results as you proceed. But in order for you to keep track of what we're doing, we're confirming all the statistics in one location.

1. Intermediate finding: If 200 million people visit YouTube daily, this suggests that over two-thirds of people watch at least one video every day. That makes sense.
2. Last but not least, if YouTube makes $10 million per day, it will make $3.65 billion annually. Once more, this is in the right order of magnitude given that Google generated over $120 billion in ad income in all regions of the world in 2018.

Solved Examples

1. Estimate Pixel Phone Storage Cost
2. Market sizing for Self Driving Cars
3. Estimate the number of Uber Drivers in San Francisco
4. Estimate the size of paint market

India demographic statistics cheat sheet

You may need to make assumptions about various demographic aspects of India for several of the estimation questions. A few of the more typical ones are listed below:

Population

Population	1,412,000,000 (2021 est.)
Bangalore Population	1.32 crores (1% of India)
Rural Population	70%
Urban Population	30%

Age Wise Divide:

0–14 years	28.6% (male 55% / female 45%)
15–64 years	63.6% (male 60% / female 40%) (2009 est.)
65 and over	5.3% (male 45% / female 55%) (2009 est.)

Education Level

Literacy in India	74%
Urban	82%
Rural	65%

Gender & Life expectancy

Male : Female	1:1
Avg Life Exp.	70 years

Income

Avg Income per individual	10000 per month

Automobiles

2 Wheelers	15 Crores
4 Wheelers	20 Crores

Mobile Users

Mobile subscribers	120 Crores
Smartphone users	75 Crores

Internet Penetration

Users	65 Crore
Urban	60%
Rural	40%

Chapter 7: App Critique

The app critique is one of the easier interviews you'll encounter when interviewing for a product manager role.

Establishing App Critique Objectives

It's important to understand how the critique will be carried out. Your interviewers might give you a general prompt like "critique app X" or get specific: "how can app X maximize revenue from its existing customer base?"

In either case, start with high-level objectives:

- **Context.** If you're bringing in an app yourself, let the interviewers know why you've selected it. If an app has been picked for you, you can set the context based on what you know about it.
- **Goals.** If the prompt is open ended, narrow the scope by establishing goals. What's the business objective? What is the user trying to achieve? What are we trying to get at in the next 20 minutes?

Think of the app critique like improve you're building on top of interviewer feedback.

Five steps for Critiquing Apps

Now the fun part. Here are six frameworks to help you examine the app:

1. **Jobs to be done.** Uncovering the core objective.
2. **Personas.** Goals, motivations, scenarios.
3. **User familiarity.** New users, intermediate users, and experts.
4. **Inclusivity.** How accessible is this app?
5. **Zooming in and out.** Aesthetic, functional, and strategic layers.

Many of these frameworks complement each other, and you'll use a combination of them in your critique.

Jobs to be done: uncovering the core objective

The job varies depending on the context. Uncovering the jobs-to-be-done, or JTBD, will provide you objective criteria to critique the app against.

- What is the customer trying to accomplish?
- If the app didn't exist, what would the customer do?
- Is the alternative better or worse than the app? Why?

A JTBD might be enough context to start getting into app flows and the UI.

Personas: goals, attitudes, and behaviors

While JTBD provides an objective high-level framework, personas can give you an extra layer of empathy with details. They're especially helpful when you're addressing users who are very different from you and the interviewer.

In the app critique context, you won't have research handy, so you'll need to make these up as you go. One way of organizing this info is through proto-personas

- **Goals.** What objective(s) are they trying to accomplish?
- **Motivation.** Why are they motivated to use the app?
- **Attitudes.** How do they feel about the task they're trying to accomplish?
- **Behaviors.** Where in the context of a user's life does the app fit in? Where do they use it—home, work, or on their commute? How do they use it? What else is happening in the background?

Pairing personas with a scenario (think mini customer journey) will help you make the most of them by putting yourself in their shoes and imagining how they would use the app to accomplish a task.

New users, intermediates, and experts

You can segment your prototypical persona further, based on how familiar they are with the app.

1. **Welcoming newcomers.** New users offer a different type of challenge compared to your everyday users who are already bought in. The app will need to communicate quickly to entice the user to continue. Depending on how much time you have, consider stepping all the way back to the initial app discovery phase. How do the users find the app? What gets them to download and try it out?

As a new user who downloaded the app:

- What is the value proposition? Can I easily find it? Does it resonate with my needs?
- Do I need to sign up or can I try it out without commitment?
- How much effort do I need to make in the beginning?
- What barriers or friction do I encounter during onboarding?
- Is it clear what I need to do next?
- How much guidance do I need in the onboarding process?
- How does this app incentivize me to use it more?

The new-user perspective might be easiest to take during the app critique. It doesn't assume prior app knowledge and it gets you and the interviewer on the same page by starting out fresh.

2. **Becoming a regular user.** With continued use of the app, the customer gains experience and becomes an intermediate user. Tasks that took a long time are a snap to complete. For an app critique, this provides a rich exploration area—how do I as a new user become a repeat customer?

- How does the app guide me?
- Where can I find shortcuts? What accelerators exist to improve my workflow?
- How does the experience grow with my needs?
- How can the experience be more personalized?

This approach works well with popular apps like Facebook that you and the interviewer use frequently. Be sure to establish some common ground first, though, as they might be proficient on parts of the app that you don't use.

3. **Expert users.** Depending on the application, experts have taken the time to squeeze every last bit of efficiency out of the tool. Similar to intermediate users, you can think about evaluating the experience from a transition perspective:

- How does the app level up intermediate users?
- Where can you find advanced functionality?
- What tools are missing that might be helpful for an expert?

For the app critique, this might be the hardest perspective to take, but it might be worthwhile for tools that are highly task oriented. Usually these tend to be enterprise apps like Photoshop or Visual Studio. Taking this perspective shows that you know the user well and that you've developed a point of view on how to meet your user's needs.

Inclusivity and accessibility

So far, we've talked about ideal user journeys and expert evaluation techniques with no mention of edge cases or customer impairments. If you want to step up in the critique, consider accessibility. Ensuring apps don't exclude users is not a nice-to-have—in many parts of the world it's the law. The Google Design team created an accessibility card deck to generate different scenarios.

The Google Design team created an accessibility card deck to generate different scenarios.

I won't be able to do justice to accessibility here, but for the purpose of the app critique you can think about the following impairments:

- **Vision.** What affordances does the app make for users with low vision, color blindness, or blindness? Think size, color, contrast, or environmental factors like screen glare.
- **Hearing.** How does the app make use of sound? Does it provide backup cues? Does it have alternative text or captions? Think about the app's use in a busy environment—for example, a busy train station.
- **Motor.** Does the app require intricate touch gestures? Are tap targets large and easy to discover? Think about using the app while walking.
- **Cognitive.** How much cognitive load does the app require of the user? Does the user need to memorize instructions or remember complicated operations?

Zooming in and out on the experience

As you're going through the customer journey, you'll at times be going back to the strategic goals of the critique to make a point, diving into app functionality or zooming all the way in to a particular micro-interaction.

One way to think about this is a top-down structure that encompasses **visible** (aesthetic) **and invisible** (functional and strategic) layers:

1. **Aesthetic.** How it looks. Domain of visual design, motion, sound, touch.
2. **Functional.** How it works. Domain of information architecture and interaction design.
3. **Strategic.** Why it exists. Domain of value props and business strategy.

Aesthetic Layer

This layer is a summation of all the parts that you experience when you open up the app. How does this application achieve visual design and consistency?

- **Brand.** What message is the app trying to convey?
- **Visual.** How does the app use the visual language (color, shape, type, and so on) to reinforce its identity? What is the aesthetic experience like?
- **Motion.** How does animation help orient the user? Are there any signature moments?
- **Sound.** How does the app use sound to inform or entertain?
- **Touch.** What gestures exist and are they easily discoverable?

Pairing this layer with personas will be helpful. This is also a prime layer to think about accessibility and inclusivity.

Functional Layer

Below aesthetics lies functionality—these are core mechanics that help the user accomplish their goal.

- **Information architecture.** How is the information labeled and organized? What paths exist to browse or search?
- **Interaction design.** How does the app work? What macro and micro flows exist? What patterns are being used, what are the trade-offs?

This layer pairs well with personas and expert evaluation.

Strategic Layer

In this layer you'll find problem framing and value proposition.

- **App.** Why does this app exist? For whom? What problem is it solving?
- **Business.** What problem is the company trying to solve? What are the values that we believe in? How does the company want to portray itself? How does it make money? What's the company's mission? What values does it abide by?

The strategy layer overlaps significantly with JTBD, so you'll probably be going back and forth between the two.

Avoiding Common App Critique Mistakes

Mistakes that designers make when doing critiques for interviews.

Shopping at aesthetics: Sometimes it's tempting to dive right in and point out all the things that look off on the UI. Don't get stuck on aesthetics. None of that will matter if the app doesn't help the user achieve its core goal in the first place. Establish your first principles based on the frameworks above and work forward.

Stuck on one approach: If you have an app that you've already picked yourself, then you've already come prepared with how you're going to critique it. But be ready to adapt your approach and consider different methods when prompted. Doing so also allows you to show off other tools at your disposal and handling ambiguity is a sign of maturity as a designer.

All praise, no substance. I've encountered situations where instead of critiquing, the designer lavishes praise on the app. As an interviewer, this tells me that you can't reflect critically on the work, and you assume that whatever's been launched is best.

Spending too much time describing the UI. Don't get stuck explaining what you see or what the app does. If you're not actively sharing your mobile app screen where the interviewer can see it, it's ok to give some context but make sure you focus more on the critique itself. Interviews want to know you're thinking; they don't want you to recite the UI.

Always ask why. Why does this work well—what makes it good? Why does this not work—what things make it less effective? A critique framework helps you stay objective and avoid this pitfall. I also recommend looking at similar apps and comparing them against

Solved Examples

1. Tips for App Critique
2. Critique Amazon App
3. Critique Google Maps
4. Critique Spotify

Chapter 8: Root Cause Analysis (RCA)

One of the most commonly asked question types in product management interviews- "Find Root Cause." Root causes test around fundamental problem solving, structuring, and narrowing skills. These are generally just one-liner questions that would create a false impression of it being easy, doable, but it is one of the most complex problems in a product interview. At the end, your approach should always converge.

Framework

On a broader level, to perform root cause analysis, breakdown the problem at every step and follow the below mentioned framework:

Clarify

Before jumping on to solve a problem, make sure that all the key terms used to define the problem are clearly understood. It might be the case that the problem conveyed by the customer/interviewer is very different from the problem that you perceived. For instance, the interviewer said that customer is not being able to view a particular feature in a product. This open-ended statement could mean a whole lot of things. It could mean that feature is not intuitively placed and thus couldn't be located by the user. It could mean the option to use that feature is not present on the interface. Which is why it is very important to get the problem properly clarified and define the key terms unequivocally. Start with asking few relevant questions to clarify the problem.

Verify the metrics defined in the problem

At the heart of any decision lie the metrics that substantiate the claims in a problem. Thus, it makes logical sense to ensure that the veracity of any metrics presented in a problem are correct. For instance, let's say that a user analytics report says that YouTube average view duration is down 20% (Month over Month) and reducing further since last month. In most situations, data provided by the customer has bad quality or they have wrongly defined threshold value that could be causing these wrong values to show up on the user analytics report. You should ask questions such as: i) how are we quantifying the Engagement? Ii) Since how long the engagement is being affected? Iii) did we change the tool which used to measure this cancellation rate? Thus, doing a quick double-check on the veracity of the metrics presented in the problem statement, would allow you to clearly see if a problem actually exists.

Check for external factors

An excellent way to test the involvement of external factors in a problem is to ask "Is this an industry-specific problem or company-specific?". Industry-specific problems can have root causes in factors beyond product line of work. External factors can be: i) New campaigns or announcements by competitors ii) New product entry in the market/product substitute iii) Larger macro-economic factors like new government regulation/policy change and iv) Sentiments on social media or significant User behavior change. For instance, if video streaming product's analytics team identifies that the number of monthly subscriptions has gone down by 30% in the last few weeks, it may well be that the drop had nothing to do with the product itself. It may be

that competitor has sliced their subscription fee causing more people to move to their product or the government could have announced a new tax on video streaming services thereby influencing people decision on whether to subscribe to such a service in the first place.

Check for the demographic being impacted

It is general good practice to understand which group of end users the problem really impacts. Is this a problem faced by all users alike or is this something faced only by a specific subset of people?

List out any recent changes or internal factors

A good place to start identifying the problem, would be to ask when the problem was first spotted? And recently is there any changes made in user flow? It would give the problem's origin and also give you a rough timeline to potentially observe patterns that could give a clue on what might have caused the problem.

User Journey

Try to make an end-to-end user journey for the product so that you can cover every tiny point of failure where a problem can reside. Any user interaction with a product can be broken down into the buckets of input, process and output. Analyzing the user-flow through each of these segmented endpoints would allow you to derive insights on the problem. A example can be, a user is complaining that he is unable to see the changes he makes to a certain segment of the web portal. Using the above segmentation process we can divide the flow into 3 buckets. The problem could lie at the place where the information is being entered (input), it could be because of the way that input is getting processed by the back end (process), or it could be that the stored information is not being displayed properly to the user on the front-end (output). Following this flow will help eliminate problem domains in a step-by-step manner.

Follow a top-down approach (Prioritization)

This is the time to identify as many causes as possible. Techniques that can be used are the 5 Whys or Fishbone analysis to narrow down the potential underlying cause or causes of the problem and the major contributing factors. To explore all potential options and to dive down into the most likely ones, you can form a tree structure with the problem statement at the root of the tree. Segment the problem into sub-problems and explore each of those sub-problems in further detail. When doing so, prioritize and determine which of these sub-branches can be ruled out completely, as and how we gather more information. Explore down the branches that are not ruled out and iteratively apply the process until the Root Cause is identified.

Solution, Recommendation, and Implementation

It is a good practice to end the discussion by giving possible solutions to the root cause. Come up with multiple solutions and prioritize them using impact and effort matrix.

Practice Question 1

You are the PM for a streaming video service. You notice that one key metric has dropped by 80%. How will you find the root cause?

Describe the feature/product

The company's primary business is its subscription-based streaming service which offers online streaming of a library of films and television programs, including those produced in-house. We offer video streaming in the US and internationally. Is this correct?

Interviewer: Yes.

Clarifying questions

Is this a new product (post launch)?

Interviewer: No, we've had this service for years.

What's the absolute number of the drop (sometimes the absolute number is small but percentage seems large)?

Interviewer: Assume that the absolute number is also large.

Does this drop affect US, international or both markets?

Interviewer: All markets.

Are internal or external stakeholders affected by the drop?

Interviewer: External.

So, the drop is affecting our end users, i.e. customers?

Interviewer: Yes

Issue is in which device, mobile or desktop?

Interviewer: Mobile.

Do we offer it on all types of phones (e.g.: iphone, android)?

Interviewer: Yes

Did the drop was device specific (e.g., iphone only)?

Interviewer: No

Let me restate the problem,

We've been providing streaming video service to customers in the US and internationally for years. The product is available on desktop and on mobile (all devices). Today we discovered an 80% drop in key metric on mobile platforms (all devices) that affects external customers in all markets

(US and internationally). I'm in charge of investigating the problem and finding the right course of actions. Is this correct?

Interviewer: Yes

As this is affecting our external users, i.e. customers, I'd like to draw the user journey and try to identify where the drop occurred. Does this sound good?

Interviewer: Yes

As a user of the video service, I download the app on mobile to watch the content in the car/etc., I create the account/pay subscription fee, log in, search through database of videos, filter the output, find the video I want to watch, watch it, log out, repeat. Is this the right sequence? Did I miss anything significant?

Interviewer: Yes, it's correct and no, you didn't miss anything.

As you mentioned the problem only affected mobile users, is % of app downloads the same?

Interviewer: Yes.

What about % of paid mobile subscribers?

Interviewer: The drop is in paid mobile subscribers.

Is it a monthly/quarterly or daily subscriber?

Interviewer: Monthly.

With 80% in paid mobile monthly subscription, I'd need to investigate further.

Internal

Did we change how we calculate this metric?

Interviewer: No

Has the drop happened in the past (seasonality)?

Interviewer: No, that's the first time the drop is that dramatic.

Was there a technical incident, bug, server issue that affected the number of downloads this month?

Interviewer: No.

Did we release a new version that's lagging technically or on UX?

Interviewer: No

Did we release a new free feature that's cannibalizing our paid service?

Interviewer: No

External

Did the % of mobile devices sold drop this year/quarter/month?

Interviewer: No

Are there any micro/macro-economic factors that affecting the decline in % of paid mobile subscribers?

Interviewer: No.

Is there a competitor that's taking our market share?

Interviewer: Bingo.

I assume the competitor is focused on both the US and international market and specifically on mobile. Is the competitor providing free service or a major discount?

Interviewer: Free video content on mobile apps.

I'll restate the root cause:

We've been providing a paid streaming video service to customers in the US and internationally for years. The product is available on desktop and on mobile (all devices). This month we've discovered an 80% drop in monthly paid subscription on mobile platforms (all devices) in all markets (US and internationally). The reason for the drop is a new competitor who entered a video mobile market and offers a free video content.

Interviewer: This sounds good.

Note: There is no one way of solving root cause analysis questions, at the end it should have a logical structural breakdown to come up with most probable root cause.

Another way to analyze such questions is by listing all the possibilities from your end and take a buy-in from interviewer at every step.

Practice Question 2

Q. 15% drop in usage of Facebook groups. Identify the root cause.

Usage definition in this case can be: User can post, User can read, User can connect, User can discuss, User can buy/sell.

Facebook groups can be formed on following three topics:

Topic Discussion: Climate Change, Political Views, Professional Views

Market places: Buy & Sell Groups, Apartment Rent

Connections: Dating, Alumni, Friends & Family

Interviewer: For the sake of simplicity, pick "Topic Discussion" as target and "number of posts" as usage metric.

Rephrasing the original problem: "Posts per Group per User" down by 15% for Topic Discussion Groups for the following types of topics:

- User Description: Active User, Motivated, Spread across Demography/Geography
- User Motivation behind post,
- Share Knowledge
- Share Viewpoint: Feel connected with peers, Feel connected to cause, Feel part of group
- Ask Help: Looking for information
- Market Product: Sell Something, Market their product or services.

Interviewer: let's concentrate on "Sharing Viewpoint" as most common motivation behind users posting content on Facebook Groups.

Looking at User Journey: User likes a topic -> User searches for a group -> User joins the group -> User reads/like/comments on posts -> User makes his first post -> Other users read/like/comment on his post -> User engages with other likeminded users.

User Pain Points (Probable):

Users don't feel a need to post anything: Concept of community doesn't exist anymore.

Users don't view groups as viable options for posting anything: They find other channels more viable, Advertisers crowd the group.

User find difficult to post something in groups: Stricter policy, UI not supportive, Privacy issues.

Users don't get what they seek by posting in groups: Other users don't engage (Most Likely), Community is getting weaker.

Interviewer: You have breakdown the problem properly, Users don't get what they seek by posting in groups is the major reason for reduction in posts.

Can you give some solutions how we can improve this?

I will redefine the problem,

Redefined Problem: Improve user engagement in Facebook groups so that users are motivated to post their view points in groups.

Solutions: I) Tweak Feed algorithm to prioritize group posts ii) Auto post most engaging posts across similar groups iii) Personalized recommendation of groups to users iv) Group Chat feed

Considering drop of 15% in usage metric, I would recommend implementing Group Chat functionality for groups. User can participate in live chat in groups. This will lead to high engagement and improved motivation for users to post content leading to improved usage metrics.

Interviewer: This is good enough.

Solved Examples

- RCA Youtube Watch Time
- RazorPay - RCA

Other Questions to try yourself

1. Order cancellation on Amazon is up by 10%; find the root cause?
2. Instagram Stories Engagement is down 30% (week over week) from last two weeks.

Chapter 9: Dashboards

Dashboard reporting helps you make well-informed decisions by allowing you to not only visualize KPIs and track performance but also interact with data directly within the dashboard to analyze trends and gain insights. Tracking metrics is crucial for product managers. Product managers require data-driven insights to illustrate the performance and growth of their products because they interact with a variety of stakeholders.

Dashboards are a brilliant solution; a visual tracking approach to selected product KPIs is a product management dashboard. It provides a real-time picture of the state of the product's life cycle.

How do you create one?

Whatever a dashboard's intended use, it should display Key Performance Indicators (KPIs) based on a company's high-level or low-level objectives and represent the specific needs and culture of that company. These KPIs will serve as metrics i.e., quantifiable measurements of each aim.
Business/Product health can be analyzed based on this list of essential product management KPIs that every product manager's dashboard should contain. Below are a few KPIs that can be included.
Refer to KPI examples for information on Industry wise KPIs.
Refer to the Metrics section of the book for more information on prioritization.

Customer KPIs

Using customer KPIs helps teams determine if customers are satisfied with the product and showcase the business value delivered to the company.

1. **Customer satisfaction score:** How many customers are happy with the product.
2. **Customer retention rate:** The average number of customers retained at the end of a specific period
3. **Conversion rate:** The percentage of customers who are purchasing the paid version of the product versus the ones using the free version
4. **Average lifetime revenue** that the company receives from one single customer
5. **Customer acquisition cost:** The average cost of acquiring customers

Product quality KPIs

Product quality KPIs establish product management goals. They give insights on improving the product and reducing the defects count.

1. **Number of support tickets a month:** Number of customers requesting product support
2. **Defect rate:** Number of defective products manufactured in a specific period

User engagement KPIs

User engagement KPIs allow teams to track customer engagement to check if users find the product suitable for their needs.

1. **Monthly active users:** Number of product users over a specific period
2. **Average session duration:** Time spent on product usage per customer
3. **Number of user actions per session:** The steps users take when using the product

Business performance KPIs

Measure business profitability by including KPIs that track the business performance of the product.

1. **Monthly recurring revenue:** The amount of revenue generated each month from one product
2. **The average revenue per user:** The total revenue generated by an average customer over a specific period

Resources: https://amplitude.com/

Practice Questions

Example 1:
Design a dashboard to track the health of the **business**. What KPIs are you proposing to use?

Example 2:
Select a **product** from our company. You are now in charge of creating a dashboard that regularly updates our team with all essential KPIs. What do you track?

Example 3:
You are the PM for a particular **feature**. What dashboard would you design to track its effectiveness? Which metrics would be used?

1. If you were the PM for Uber, what dashboard would you build to track the health of the app?
2. You are the PM for Flipkart's cart page. How would you build a dashboard to measure success?
3. You are the product manager for Lyft Direct Debit Card for drivers. The product just launched. How do you build this product's first dashboard?

Chapter 10: Behavioral Questions

Imp: One of the most skipped sections while preparing for the interview. But there would be a couple of questions in at least one round of each interview you appear so be prepared in advanced to answer. Also please see the note below:

Obviously, we will not be giving you a model answer, because we are sure you will start copying it in all your interviews!! Trust us we have all been there 😊. These are just guidelines that gives you a direction. Remember each one has a unique story so do leverage it when you get the chance.

STAR Framework

S - Situation
T - Task
A - Action
R – Result

These are not just a bunch of random words combined to make a word. This is a framework which is typically followed by interviewees to answer behavioral questions in a structured manner. The key is to quote your real-life examples.

Example:
"Describe a situation when you had to communicate complex information to your team"

Answer:

- **Situation** – Describe the situation in detail. What were you working as? Your role? Did you have someone who reports to you? Did you have to give this complex information to your boss too? What was that 'complex' information? Describing the whole situation to your interviewer as a story helps him/her visualize it and sets the context well.
- **Task** – Now, when you were in that situation, what were you expected to do? What were the challenges that you were expecting? How were you planning to solve them? What exactly was your plan? What was the final goal? Did you have to convert the complex information into something simpler?
- **Action** – After all the planning of tasks, how did you finally execute it? Any unexpected challenges during the execution? What went right and what went wrong? What tools did you use?
- **Result** – This is the easiest part because you would obviously choose a situation when the result was on your favor. So just say it with confidence! How did it help you and all the stakeholders involved and the company.

"As a part of the project, I had to report to my team on my findings from the qualitative and quantitative research into customer trends. The information consisted of so much data which was very unorganized for someone to understand intuitively. I had to find a way to make this information simpler. I had to analyze the information, plot data as graphs for better understanding. I made use of my Excel and Tableau skills to present data in the simplest way possible. My presentation was well received, and the team also started looking at market research in a new light."

SOAR Framework

S – Situation
O – Obstacles
A – Actions
R – Result

Like STAR framework except that we talk about 'Obstacles' here. After explaining the situation and while you explain your approach to the problem, you could talk about the obstacles/challenges you faced and how you resolved them.

CAR Framework

C – Challenge
A – Actions
R – Results

- **Challenge** - Describe a situation or a task that you needed to accomplish and where there were some difficulties to overcome. Describe a specific event or situation. Give enough detail for the interviewer to understand.
- **Actions** - Describe the action you took and keep the focus on YOU. Even if you were part of a group, describe what YOU did. Tell what you DID, not what you would do.
- **Result** - What happened? How did the event or situation end? What did you accomplish or learn?

Note: All the three frameworks mentioned above are extremely useful to answer questions like "Describe a situation when....", "Tell me about a time when...." where the interviewer wants to understand your approach to the problems you had been facing at your work, college etc.

Questions to Prepare

Why Product Management?

You should be well aware of the roles and responsibilities of a Product Manager. The challenge here is that PM role is not standard across all the organizations. Hence, you should do thorough

research into the role at the company that you're interviewing for. Link the role's qualities with yours and let them know how you think you'll be the right fit. Back the qualities up with examples. For example, if you say that you're a visionary person you must quote an example from your life when your vision helped you achieve something.

What if you do not get into Product Management?

Never give answers like "I don't know", "I haven't decided" etc. Planning tasks is one of the key roles a PM holds. It is always a good thing to let them know your plan B because even they know that it is not possible to always get everything right at the first attempt. You could frame an answer in which you tell them that you will learn from the mistakes and keep trying. You could also tell what other roles interest your apart from the PM role (Eg. Marketing) and tell them how you would go about trying for a different role.

You do not have relevant work experience (IT), so why do you think you are fit for this role?

It is one of the toughest questions that most freshers face in an interview setting. One suggestion would be to participate in as many PM case competitions as possible. There is a lot of attention for winners or finalists of national level case competitions. Even if you do not have any such achievements, you should let them know how keen you are to learn and get better at it. You might talk about any PM related courses/certifications (Udemy or Coursera) you have done. As it has already been said, it is always good to know the roles and responsibilities and to link it with your qualities.

Tell me a challenge you faced while working as a team? How did you overcome it?

This answer varies from person to person based on their personal experiences. Freshers might not have any examples to quote from the workplace, but you can always talk about a time when you worked in group projects and the challenges you faced and how you resolved them. Don't forget to mention the learning from it. If time permits, also mention how you applied those learnings later in life and what came out of it.

Delivering faster vs. Delivering a better product. What would you choose?

It is one of the most common trade-offs. Don't give a binary answer because that is never true in a real-life situation. Talk about the situations in which you would prefer one to the other. The answer always depends on what kind of a situation you're posed with. Either ask the interviewer to describe the situation or you yourself come up with a situation and explain how one approach is better than the other in that situation.

Tell us about a time you used data to influence an important stakeholder.

Data has become a vital part of every business. All the product management interviews involve test on your number crunching and analytical skills. It is an added advantage for you to know tools like Tableau, PowerBI etc.

Why do you want to work at our company?

It is important to do thorough research into the company. This is not just for the PM roles but also for any interview you sit for. The company doesn't want to hire people who do not fit in with their culture and vision. So, understand the company's values, vision, mission, culture, products, competitors etc. And talk passionately about those in any way you can and tell them this is why you want to work for them.

What's your approach to prioritizing tasks?

Prioritizing what features to work on is one of the important tasks a PM has to do. This is to make sure the development team isn't overloaded with unnecessary tasks and also the important features get delivered on time. You will understand how to do it when you read other PM-related material during your prep. Just list the step-by-step process and clearly tell them.

What main changes would you make to our product?

It depends on the company. Your research into the company helps here. The key is to talk from the user's perspective while keeping financial and technical feasibilities in mind.

Note: All the questions mentioned above are not exhaustive. There can always be more questions an interviewer might ask based on your previous answers. You should always be ready to defend your answers and to justify your resume points.
The other kind of questions which might come in an interview (not necessarily related to Prod Man) are as follows to name a few.

- Tell me about yourself
- Strengths and weaknesses
- Are you okay with accepting your fault whenever you make a mistake?
- Do you think you are a good cultural fit?
- Tell me something that's unique to you
- Hobbies and interests

Advanced PM Topics - Optional Topics

These are advanced topics for preparing for Product Management – **Chapter 11 to Chapter 15** which are Project Execution, Wireframing, Company Frameworks, Product strategy and System Design. This is something which will not be asked in your summer interviews in most of the companies. You can look at these topics while preparing for the final placements. These topics would be useful to go through and apply in your day to day work while interning in your summers

Chapter 11: Project Execution

AGILE Framework

Agile is the ability to create and respond to change. It is a way of dealing with, and ultimately succeeding in, an uncertain and turbulent environment.
Agile is an iterative approach to project management and software development that helps teams deliver value to their customers faster and with fewer headaches. Instead of betting everything on a "big bang" launch, an agile team delivers work in small, but consumable, increments. Requirements, plans, and results are evaluated continuously so teams have a natural mechanism for responding to change quickly. Whereas the traditional "waterfall" approach has one discipline contribute to the project, then "throw it over the wall" to the next contributor, agile calls for collaborative cross-functional teams. Open communication, collaboration, adaptation, and trust amongst team members are at the heart of agile. Although the project lead or product owner typically prioritizes the work to be delivered, the team takes the lead on deciding how the work will get done, self-organizing around granular tasks and assignments.
Agile isn't defined by a set of ceremonies or specific development techniques. Rather, agile is a group of methodologies that demonstrate a commitment to tight feedback cycles and continuous improvement.



Agile principles

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. Business people and developers must work together daily throughout the project.
5. Build projects around motivated individuals. Give them the environment and support they need and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity–the art of maximizing the amount of work not done–is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Agile methodologies and frameworks

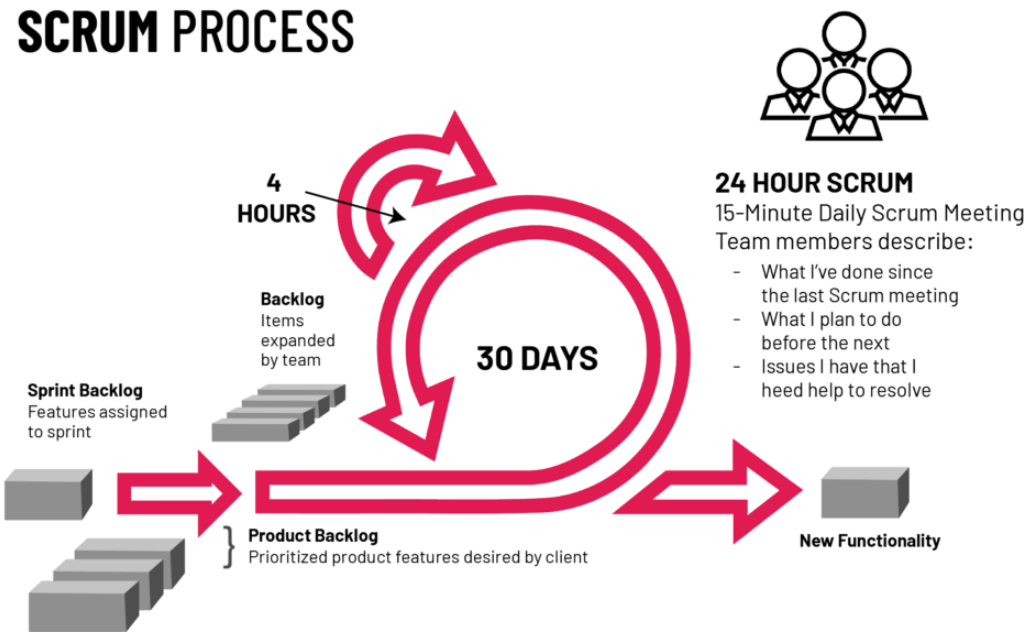
An agile framework can be defined as a specific software-development approach based on the agile philosophy articulated in the Agile Manifesto. Various organizations use many popular frameworks. Often these organizations modify parts of the frameworks as they see fit and iterate on their own agile processes. There are several Agile methodologies and frameworks, each with its own pros and cons. Some are hybrids of multiple methodologies. Scrum is by far the most commonly used Agile methodology; among other being Kanban, Lean, Extreme programming, Feature driven development, etc.

Scrum

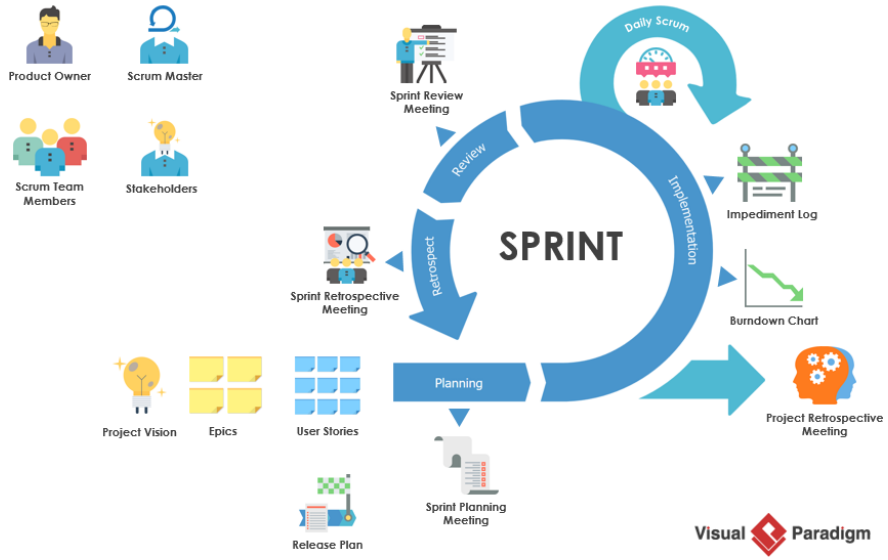
It is a lightweight process framework for agile development, and the most widely used one.
A "process framework" is a particular set of practices that must be followed for a process to be consistent with the framework. (For example, the Scrum process framework requires the use of development cycles called Sprints, the XP framework requires pair programming, and so forth.)
"Lightweight" means that the overhead of the process is kept as small as possible, to maximize the amount of productive time available for getting useful work done.
A Scrum process is distinguished from other agile processes by specific concepts and practices, divided into the three categories of Roles, Artifacts, and Time Boxes. Scrum is most often used to manage complex software and product development, using iterative and incremental practices. Scrum significantly increases productivity and reduces time to benefits relative to classic "waterfall" processes. Scrum processes enable organizations to adjust smoothly to rapidly changing requirements, and produce a product that meets evolving business goals.

An agile Scrum process benefits the organization by helping it to

- Increase the quality of the deliverables
- Cope better with change (and expect the changes)
- Provide better estimates while spending less time creating them
- Be more in control of the project schedule and state



The Agile – Scrum Framework



A/B testing

An A/B test aims to compare the performance of two items or variations against one another. In product management, A/B tests are often used to identify the best-performing option. A/B testing involves splitting your audience/user base into two groups.

The two groups then receive two versions of a feature. It could be something as small as a different call to action button color, or it could be a re-ordering of your onboarding sequence. After a certain amount of time (which will differ depending on the test/product/goal etc.) a product manager will look at the data gathered to better understand which version should be rolled out to the entire user base or learn which of those versions' users respond to most favorably.



Although it has historically been primarily a tool of marketing and advertising, A/B testing can also help product managers build better products.

Product Roadmap

A roadmap is your vision for how a product (or product line) will help achieve your organization's strategic goals. A good roadmap inspires. It inspires buy-in from executives, inspires confidence from customers and salespeople, and inspires development teams to produce the groundbreaking products that drive significant growth.

The litmus test for a good product roadmap is that it's **visual**, **accessible,** and **clear** enough for anyone to scan for answers to the following questions:

- What are we doing?
- Why are we doing it?
- How does this tie back to our OKRs?

How to build a theme-based product roadmap?

A theme-based roadmap replaces release dates with time horizons and generally defined in three columns:

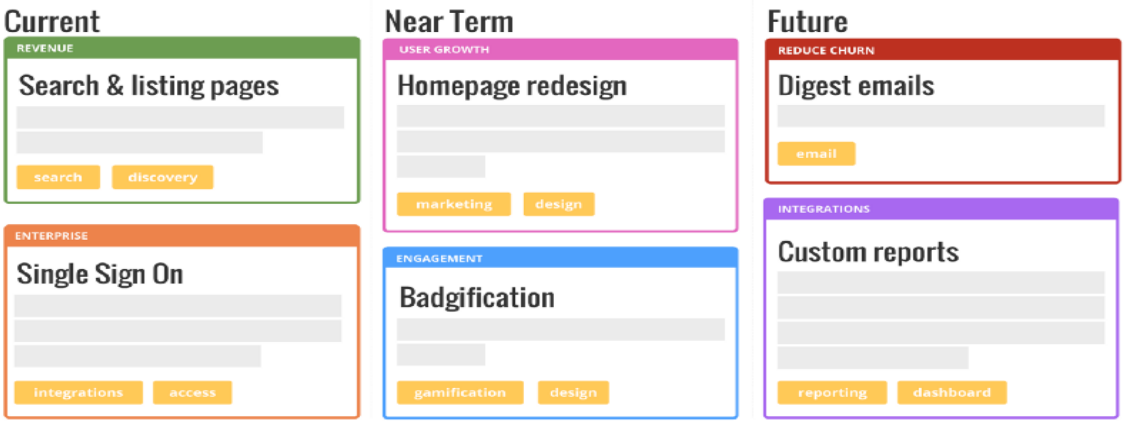
- **Current:** Stuff you're currently working on
- **Near term:** Stuff that are coming up soon/next
- **Future:** Stuff that you'd like to work on in the future, but need to do a bit more research

Roadmap is a bird's eye view of priorities and doesn't show any dates unlike release planner. This allows flexibility to accommodate changes across time horizons.

A theme-based roadmap incorporates broader themes instead of individual features, working on themes gives you the space to explore how you might want to approach each problem, and decide how to get it done later.
Added themes in the roadmap are then attached with further details answering, *what are we doing, why are we doing and how does this tie back to our OKRs*. These details generally include:

- **Ideas:** Tactical suggestions for improvement. These ideas answer a simple business case: *What problem are you trying to solve?*
- **Customer feedback:** We attach feedback directly to ideas, so they can be linked to potential improvements, and we can easily track what our clients are asking for.
- **User stories:** Use case scenarios for ideas: As a user, I want to X in order to Y.

Cards sitting on the Future column wouldn't have all those answers, but as a card moves closer to Current term, they should become a lot more detailed. These cards later tagged and color coded for better visibility and communication.



Go-to-market (GTM) Strategy

A go-to-market (GTM) strategy is a plan that helps you define your ideal customers, coordinate your messaging, and position your product for launch. A GTM strategy also keeps key business units aligned on the same plan, allowing you to meet a market need and effectively iterate on your product.
A GTM strategy is needed when launching a new product in an existing market, launching an existing product in a new market, or testing a new product's market for growth. It allows companies to effectively position themselves against competitors, create scalable inbound and outbound models, and leverage appropriate tactics to achieve goals.
There are two main types of GTM strategies:

Product Led GTM Strategy

It uses the product itself to acquire and retain users. The product serves as a salesperson by providing so much value, the user can't help but upgrade their package. Calendly and Slack are great examples of product-led growth in action.

Sales Led GTM Strategy

It uses marketing to drum up interest for a product, capturing it in content and demo forms. Salespeople then reach out to those prospects, with the goal of converting them into customers.

Stages in building a GTM strategy

- **Defining your Ideal Customer Profile (ICP) –** It is a customer who can find massive benefits from the product or service and gives enough value in return to make business profitable
- **Researching competitors –** Understanding where your product or service fits in the existing landscape is key to any GTM strategy
- **Developing your messaging –** Communicating the value of your product or service to your ICP in a way that resonates with their pain points. And to be able to do this, you need to speak their language
- **Setting targets –** Having clear model is crucial, this allows to build full-funnel pipeline and map out data-driven targets and sales goals which reps work towards each month
- **Choosing your tactics –** To reach your ICP leveraging data, marketing strategy, content plan and strategic partnerships forming unique tactics to follow
- **Providing feedback –** Establishing a feedback loop between marketing, sales and product development will make sure the key learnings from the GTM strategy are actioned



Note: For other frameworks please refer to STP and 4P which is explained in the marketing compendium. Also you can look at cases of Market Entry frameworks from Consulting resources.

Product Requirements Document (PRD)

Product Requirements Document (PRD) is an initial document which describes about your problem, market opportunity, whom you are targeting, product used to solve that problem, potential stakeholders involved, order of tasks to be followed, and how to measure the performance of your product. The PRD enables collaboration between different teams in the organization. It is the job of the product manager to take inputs from the customer, engineering team, design team and sales team and keep them on the same page.

What to include in a PRD ?

This is not an exhaustive list and may vary from project to project.
Your PRD should always follow a cohesive story from the beginning to the end and each section should be connected to the previous and the next section

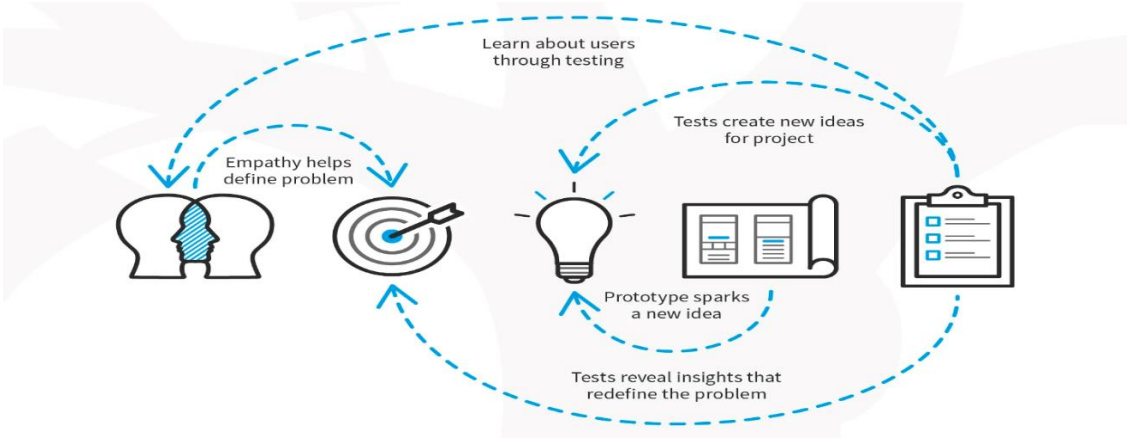
- **Problem definition** - This would include a brief description of the problem at hand and it can highlight any gaps or ungrounded assumptions that you are making. State them out early so that there is no confusion later on.
- **Market Research/Opportunity** - This section would include research on competitors, opportunity in the market and market sizing if applicable.
- **User research** - Using primary and secondary research user demographic, behavior and pain points are identified. Surveys, focus group interviews and in-depth interviews are some the research methods that can be utilized. Key insights should be derived from the research.
- **User persona and their use cases** - Key insights from the user research will help in narrowing down the people who are facing the problem defined. A description of the persona with their use cases should be covered under this topic
- **Prioritization of the use cases** - Prioritization methods such as an empathy map can be used to prioritize the persona. One of the use cases from the persona is chosen to work upon and a reason backed by data should be given as to why the use case is being prioritized.
- **Proposed solutions** - This would include all the features that your main product will have. Functional and non-functional requirements
- **Prioritization of solutions** - Impact and effort can be calculated to prioritize the solutions or the KANO model, MOSCOW model can be used if they are fitting for your particular product. Don't try to force fit any of the existing models and think what works best for your own solution.
- **UX design** - Figma and Balsamiq wireframes can be included to incorporate the solution
- **Roadmap** - A roadmap for the product including the discussions from various teams is incorporated. Different steps are:-

  Step 1 – Idea Sourcing
  Step 2 – Idea Screening
  Step 3 – Market Research & User Research
  Step 4 – Strategy Development
  Step 5 – Product Creation.
  Step 6 – Testing & Feedback Gathering
  Step 7 – Constant Product Improvement

- **Metrics**:- North star metric to be tracked should be included with a specific reasoning as to why this is considered the north star metric. App specific metrics should be included here, and you can use the AARRR framework for the same. You should also include the feature specific metrics for your prioritized solution.

Design thinking Method

Design thinking is a philosophy and a set of tools to help you solve problems creatively. Design thinking focuses specifically on the human-centered side of problem-solving. It is about understanding whom you are designing for and what their needs are and then helping you to innovate based on those needs.



There are 5 main steps to any design thinking process.

1. **Empathize** – Gives a vital foundation for creative thinking and understanding the target market for the product or service you're aiming to create. Most of the time, it involves gathering data, conducting customer interviews, and developing personas to identify their wants, needs, and goals. In this step, you are encouraged to set aside any assumptions and collect authentic insights from the end user.
2. **Define** – In the define step, you take everything you learned in step 1, start making sense of it, and define the problem. What are the difficulties and challenges users face while accomplishing a specific task? Are there any patterns? Can you see the bigger problem? By the end of the design phase, you should have a clear problem statement.
3. **Ideate** – In this step, you are coming up with the potential solution for the problem defined in step two. You are trying to match your product or service with the insights you got from empathizing and defining activity. This step may look like a brainstorming session, where all the creativity happens. You will select a few ideas to proceed with after the ideation session.
4. **Prototype** – Here, you're narrowing the concept down to a select few that you believe would be worth developing into something more realistic. You are creating a straightforward, testable prototype using these solutions. A prototype is a scaled-down

product version including probable solutions recognized in the earlier phases. This stage is crucial to test each idea and identify limitations and problems. The suggested solutions may be approved, enhanced, altered, or discarded during the prototype stage, depending on how well they fare in prototype form.

5. **Test**- User testing follows prototyping, and it is crucial to remember that this is seldom the last step in the Design Thinking process. The outcomes of the testing step will frequently direct you to a prior stage and provide the insights you require to rephrase the initial issue statement or generate novel ideas you had not previously considered.

Solved Examples

The questions below take reference of core concepts like RCA Chapter discussed earlier for solving. Please be equipped with that.

1. Failing Metrices
2. Approach to a Product Execution Question
3. Youtube Goals
4. Facebook Event Success

Chapter 12: Wireframing

UX design

User experience (UX) refers to any interaction a user has with a product or service. UX design considers each element that shapes this experience, how it makes the user feel, and how easy it is for the user to accomplish their desired tasks.

This could be anything from how a physical product feels in your hand, to how straightforward the checkout process is when buying something online. The goal of UX design is to create easy, efficient, relevant, and all-round pleasant experiences for the user.

UI design

UI refers to the actual interface of a product; the visual design of the screens a user navigates through when using a mobile app, or the buttons they click when browsing a website.

UI design is concerned with all the visual and interactive elements of a product interface, covering everything from typography and colour palettes to animations and navigational touch points (such as buttons and scrollbars).

Read more about the difference here:
https://careerfoundry.com/en/blog/ux-design/the-difference-between-ux-and-ui-design-a-laymans-guide/

What is wireframing?

A wireframe is a schematic, a blueprint, useful to help you and your programmers and designers think and communicate about the **structure** of the software/website/app/product you're building. The designs you build are also called wires, mockups, or mocks.

It is a low-fidelity mockup that serves 3 simple and exact purposes:

1. It presents the information that will be displayed on the page
2. It gives an outline of structure and layout of the page
3. It conveys the overall direction and description of the user interface

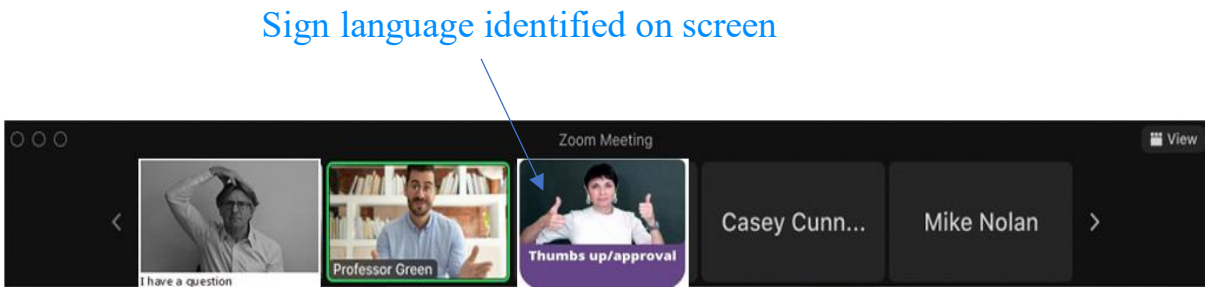
The key to a good wireframe is simplicity. They range from boxes and lines sketched on paper to onscreen creations that have a more polished look. Interactivity is at a minimum, so users can test behaviour without as much concern for a product's look and feel.

What is prototyping?

A prototype is a barebones, relatively simple working model of an app or webpage. They are typically the next step in the product design process after wireframing. Prototypes typically have color, animations, and the actual content that will be on or in your product.

Unlike wireframes, prototypes are often medium to high fidelity. They allow the user to test a digital product's interface and interactions, and this level of functionality can be useful during the usability testing of an application.

Ex: For a feature which recognizes sign language on zoom which is built for people who can't talk. The prototype may look like this.



What are the tools we can use to make a mockup?

1. **Balsamiq:** It focuses on the basics and doesn't have many features to use and therefore it is easy to pickup. They also provide a 30-day free trial and enable exporting of files once you are done with wireframing.
   Tutorial :- https://www.youtube.com/watch?v=KnZrypOaVCg&t=3s

2. **Figma:** Figma provides pre made templates which makes making a wireframe a breeze. It also provides a free version which covers all the basic tools required to create a professional wireframe.
   Template link :- https://www.youtube.com/watch?v=c9Wg6Cb_YlU

3. **Adobe XD**: Adobe XD is a vector-based UI and UX design tool and it can be used to design anything from smartwatch apps, mobile phone apps to fully fledged websites.
   Tutorial :- https://www.youtube.com/watch?v=UgKmn08-j9Q

Solved Examples

1. https://www.youtube.com/watch?v=mZwPWZP8re0,
2. https://www.youtube.com/watch?v=kk6650gC4R0

Chapter 13: System Design

What is System Design?

System Design entails creating the architecture, modules & interfaces of any product, meeting all the specifications required for it perform all features intended for the user. It provides a high level design of a software/hardware product before the implementation can be done.

Why is System Design important?

System Design is important because –

- Helps to get buy-in from stakeholders before starting the implementation
- Help different teams understand their contribution towards different facets of the product implementation
- Enables tracking progress of the product implementation
- Leads to increased scale & innovation as team-members can view different parts of the system & contribute towards its improvement

Companies asking system design questions

Meta, Google & Amazon Product Management Interviews among many others

Framework with Practice Example

There are 4 steps to designing a good system. Keep in mind to tailor your solution such that it fulfills the requirements & features given by the interviewer.

Step 1 : Understand the problem & define the scope
Don't jump to the solution. Think clearly, deeply and ask meaningful questions to clarify the requirements & assumptions surrounding the problem given.
In case the interviewer asks you to make your own assumptions (most cases they will) clearly write them down in a corner of the whiteboard/paper. These assumptions will form the base your solution.
Example of questions that can/should be asked –

- What features do we need to incorporate into the product?
- What is our target user base?
- How many users are we targeting?
- How fast are we planning to scale up the product? (Assume fixed timelines and understand the anticipated growth)
- What are the core technological competencies of the company? Can they be leveraged in this product?

Example of an interview scenario – You are asked to design a social media app.

Try to visualize any social media app in your mind. Once you have the picture of a generic social media app in your mind, ask questions to clarify how this specific app can be different/similar. Good and precise questions to be asked revolve around –

- Platform – Web/mobile?
- Features – What primary features are expected?
- Visualizations – Specific ordering of social media posts on the app? Will there be images/texts?
- User base – Target user & approx. traffic?
- Assumptions – Given login is necessary in most social media apps, I assume it will be necessary here too?

Step 2 : Propose high-level design and get buy-in from all stakeholders

We aim to collaborate with the interviewer to reach an agreement on the high-level design of the product.

For every key feature of the system, draw a high-level design.

- Come up with initial blueprint of the design. Take feedback.
- Draw high-level architecture diagram such as box diagrams depicting different modules of the system. This might include APIs, web servers, clients, data stores, cache, CDN, etc
- Do back of the envelope calculations to check if system is scalable. Think out loud

Example

Continuing with the previous example, Design a social media app, we design 2 high level designs corresponding to its 2 key features –

- **Publishing a post** – When a user posts content, the data is written into the cache/DB and the post will be populated in their friend's homepage
- **View posts** – The posts belonging to user's friends will be visible here

The 2 diagrams are –

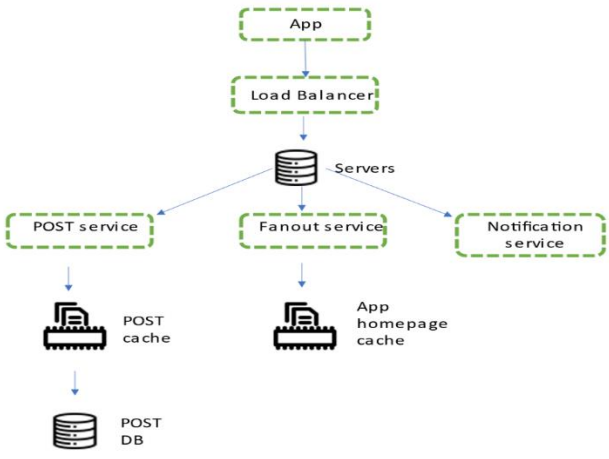


Diagram : Publishing a post

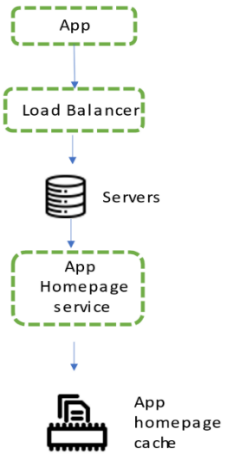


Diagram : View Posts

Step 3 : Design Deep Dive

By this point, you should have completed –

a. Converged on an understanding of the goals & scope
b. Designed a high-level diagram of the overall system
c. Obtained feedback on the design
d. Based on feedback, understood areas where you need to deep dive

Work with the interviewer to prioritize components parts of the architecture. Take feedback from interviewer to understand areas which he/she wants to know more of. Also, time management is necessary so keep the design precise and avoid unnecessary details.

Example

Continuing on the previous example, "Design a social media app", we will further emphasize the low-level details in the design.

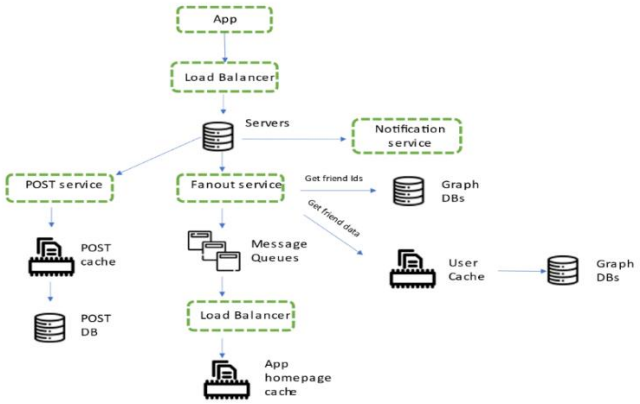


Diagram : Publishing a post

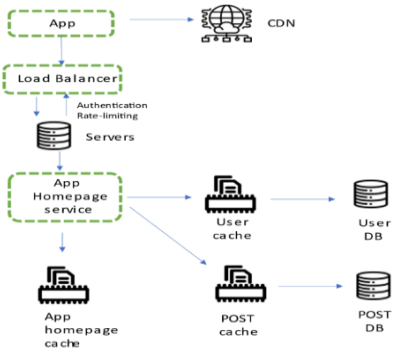


Diagram : View Posts

Step 4 : Wrap Up

In the final step, you might be asked follow up questions regarding the feasibility, scalability, performance and other aspects of the designed product.

You can close the discussion with the following –

- Identify bottlenecks & suggest further improvements
- Recap the design and Identify error cases and corner cases
- Operational & maintenance strategies for your system
- How do you plan to scale up for future releases? Any other refinements ?

Solved Examples

1. Design Twitter
2. Design Instagram
3. Design Prime Video
4. Design Web Crawler
5. Design Whatsapp

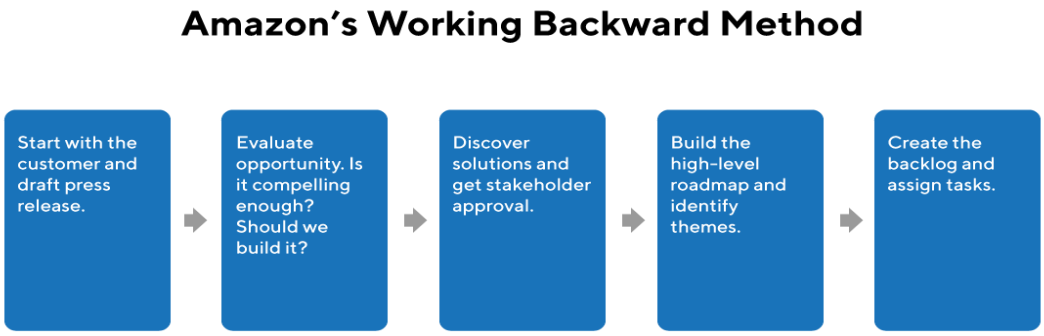
Chapter 14: Company Frameworks

The role of a product manager varies across different companies and each one defines their own job responsibilities in a way. In such a case, each company has designed a set of frameworks that aligns best with their strategy. Among product companies, we evaluate the most popular frameworks and enlist possible scenarios to use the same. These frameworks can be used as a reference, however relying too much on frameworks can hinder creative thinking.

Working Backwards by Amazon

This is unique framework used by product managers at Amazon that ensures only the best solutions reach the day of the light. Before proposing a new feature or a product, Amazon requires its product managers to frame a press release about the same. The press release must clearly articulate what problem the product intends to solve, how Amazon will solve the problem and how it will beat the existing solution.

At Amazon, it is believed that if a product manager cannot clearly articulate how his/her product will perform after its launch; the product is not worth envisioning. In fact, most managers go through several iterations of the press release before finalizing it.



Reasons for using Amazon's Working Backwards Framework

- **Alignment with Customer Focus**: The core job of a product manager is to represent the customer's needs at an organization. Developing a press release makes the product manager think from the customer's perspective and ensures that the product should work best for the customer in consideration.
- **Check for Product Feasibility**: Envisioning how the product will pan out once launched helps the product manager to validate the feasibility of the solution as well. It gives chances to the product manager to refine the product and check its viability post launch.

Product Excellence Framework by Google

This tech giant has built some principles for its product managers to follow while designing all products. Their principles act as a reference for the team to communicate with other stakeholders such as UXR and Engineering. Google believes that their product managers should be 100% sure of the products they propose to build which is possible when the follow the below principles:

- **Know your users**: A lot of product managers tend to think intuitively about a problem due to their inherent biases and assumptions. This principle advocates the PM to analyze, observe and ask users about their behaviors before finalizing on the problem to solve. Google focuses highly on primary research and believes in observing users' behavior in their natural environment.
- **Critical User Journeys**: Google describes a user journey as the set of necessary steps that a user performs to complete a task. It is based on the goal of the user and his/her identity. CUJs form an important part of customer personas where the PMs are encouraged to envision a customer performing a certain task.
- **Prioritization**: A lot of times, one thinks that there are too many important problems to solve for the customer. Google encourages its PMs to think about the most pressing and urgent need of the user and build a problem statement around that. In these cases, it becomes essential to prioritize different use cases which solve an important problem universally.
- **Pitfalls**: As a PM, one should not only know what will work but also be aware of what may fail. Awareness should be followed with preparation; a PM needs to plan for different edge cases and evaluate trade-offs to reach the final solution.
- **Product Excellence Principles**: Google defines three principles to achieve product excellence – focused utility, simple design and crafted execution. The first principle reiterates the focus on customer and its user journey. The other two principles are focused on the process of product development and how it should solve the problem in an elegant way.
- **Measurement**: Google encourages its PMs to measure and track both high-level goals and low-level metrics to continuously understand the product performance. It is advisable to understand whether the customer problem is being solved. For consumer products, one can use the HEART framework: Happiness, Engagement, Adoption, Retention and Task success. For business products, try SUPER: Serviceability & System Health, User sentiment, Productivity, Engagement, and Revenue.
- **Bringing it all together**: PMs at Google work closely with UX and Engineering teams to launch the final product and iterate through various versions. In their own words," A minimum product that delights our users can improve across dimensions when you iterate."

Reasons for using Google's Product Excellence Framework

- **Holistic Perspective**: This framework enables a PM to keep customer at its focus and think about the product holistically, considering various parameters like measurement, pitfalls, execution, design, etc. A product manager can refer to this framework starting from ideation to validation to execution and finally for launch.

- **Common Reference for All**: Following such an extensive framework can help the product manager design documentation that will act as a single source of truth for all stakeholders including UX, PM and Engineering.

Experimentation Framework by Spotify

This company believes in the experimental approach of iterating and delivering a great product experience. Product managers at Spotify are encouraged to achieve such a product experience at lower design costs through projects that are less risky. They strongly believe that products that operate at scale cannot afford to present poor experiences to users since their satisfaction can drop almost instantly. Their team follows four steps to create any product at scale:

- **Think It:** PMs at Spotify are motivated to focus a lot of user research and discover the most important problems to solve. After coming up with multiple problems, they usually go one step ahead & validate these problems through experimentation. This lowers the risk of failure, by using experimentation results to estimate the probability of success.
- **Build It:** Teams at Spotify focus on building a minimum viable product (MVP) post the discovery stage. Here, they try to build features that are necessary to solve the problem at hand and test this MVP with a set of users to gather initial feedback. At this stage, they perform multiple iterations of the product & continuously improve performance.
- **Ship It:** PMs at Spotify believe in shipping the product in phases. They release the features to a limited set of users to begin with. Based on their feedback and experience, it depends on the success of the feature as to whether they'll roll it to the entire user base. In case it does not meet they reiterate product design and not launch it.
- **Tweak It:** During this stage, the teams actively evaluate improvements in the products, and fix any bugs or introduce minor changes in upcoming versions. Their focus is to keep reducing costs and improve performance currently.

Reasons for using Spotify's Experimentation Framework

- **Fail fast, fail cheap**: Due to the iterative nature of this framework, it allows product manager to make changes to their products and test it in the market as quickly as possible without fully launching it. It allows product managers to experiment with their ideas and pivot when required.
- **Optimized Solution**: Since the team continuously works to reduce risk of failure in the market and operate at minimum costs, it offers dual optimization. The customer is ensured to get the best product experience & the product design ensures minimum costs.

Chapter 15: Product Strategy

Monetary Strategy

Pointers to keep in Mind

- Thinking Long Term
- Being Creative
- Complementing User Convenience during its journey

Prevalent Models

- **Freemium** – Having Right mix of features which you keep free, and you can charge for. Ensuring that users do come to the app to enjoy free features which serve essential needs, but highly engaged ones are ready to pay for getting those delight features
- **Advertising** – Ensuring that the user experience is not hampered due to the in-app ads. Also deciding which are the relevant ads you show in your app to have a higher Click Through Rate (CTR)
- **Subscription** – How to price? Is it valuable enough for both the company and the customer to pay $X/ month? How do we differentiate ourselves from competitors
- **Bundling** – Should we bundle this with existing products/offerings? Which version of the product to add these new features to?
- **In App Purchases** – Mostly used in games, where the basic app is free but add-ons are chargeable
- **Fixed Pricing** – One time purchase, lifetime access
- **Razor Blade** – Selling hardware or platform at a lower cost, but the apps/software's at higher prices. *Like in gaming consoles*

Pricing Methodology

Note: You can refer to consulting cases on pricing to get more idea on how to tackle these questions. But these are normally the steps you go for while deciding on how to price a product:

- Identify the business goal – market share or profitability
- Identify the trend of the industry and the phase of product in its lifecycle – early stage, growth, maturity
- Perform Competitive Benchmarking to see can the differentiated products be highly priced – offensive pricing or defensive pricing
- Readiness to pay of the target segment
- Promotional Discounts needed – initial customers, time bound, referral based
- Models
    - Cost Based – Would be easy if you have done Managerial Accounting
        - Fixed & Variable Cost
        - Determine the markup %
        - Find Breakeven quantity to set market entrytargets
    - Value Based
        - How much is the total value or benefits that the customer gets?

- Opportunity costs
- Time Value of Money

- How much is the customer ready to pay?
  - Target Segment Income and Savings
  - Target Segment Need/Desire

- Competitor Based
  - Find the price of nearest competitor and charge something similar
  - Can we go for a price war?

Typically, we go for a combination of all 3 in the interview to get a good estimate. Sample Example walkthrough: https://www.productmanagementexercises.com/1961/how-would-you-decide-on-the-price-of-amazon-prime (8 min read)

Market Entry

Note: You can refer to consulting cases on market entry to get more idea on how to tackle these questions. But these are normally the steps you go for while deciding on how to price a product:

Should we enter?

- **Company**
  - Identify the goal of the product/company? The vision of the company ?
  - What are the core capabilities of the company?
    - *VRIN Analysis*
    - *SWOT Analysis*
  - Where do we excel as compared to others?
- **Industry Structure**
  - How big is the market size?
  - What is the possible market share?
  - What is the market growth/industry trends?
    - *Porter's 5 forces*
- **Competitor Analysis**
  - How many players are there?
  - What our competitors are trying to do?
- **Feasibility Analysis**
  - Operational – Value chain, External Regulations
  - Financial - Profitability, Break Even Point, Pricing, Costs
  - Distribution Challenges – Production, Marketing, Supplier Access
  - Expertise – Is there anything that we can leverage from our existing capabilities?
- **Customers**
  - Who is our target customer? Is there an overlap with out existing competitors ?
  - Other attractive customer segments?
  - What is the value offering to the customer?
- **Alternative strategies**
  - Evaluate trade offs with respect to cost , time to market

- **Risks and Challenges**
  - Legal/Compliance Risk
  - Market Risk
  - Operational Risk

Final Decision: If we should enter or not based on the above factors

How should we enter?

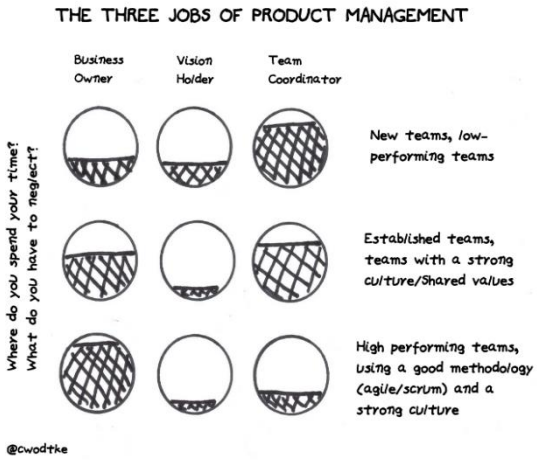
- Is timing of any importance? How we can leverage the timing of entry?
- GTM Strategy
  - From Scratch – STP, 4P (Refer to the section of GTM Strategy in Product Execution of this book)
  - Joint Venture with some other company
  - M & A with another player

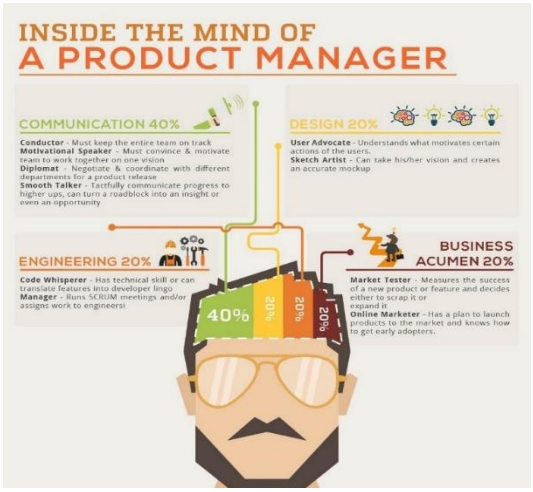
Solved Examples

1. Market Entry - Should Google enter the OTT market
2. Increase Revenue - Increase Airbnb Bookings
3. Strategy Determination – Should Facebook Consolidate all its messaging apps
4. Monetary Strategy – Pricing Spotify Premium

Conclusion

Wow! After what seems like ages, we have reached the conclusion of this book. We hope that through the gyan given here, you have gotten a more clear picture of what goes on inside the mind of a product manager. Product management is an interesting role and depending on the company you are going to work in, the role will change scope.





And by any chance were you a developer or a coder before this? Didn't you hate those times when some guy just walked up to you and ordered you to design software that didn't make sense? Guess what. Once you are a PM, you will have a chance to do that. You can walk up to a random engineer and tell them to do technically unfeasible things.



With this, we come to an end of the PM Prep Book.
Wishing you all the luck with preparation, and hope the information passed here was useful!

Resources

Videos

1. Mock PM Interviews
- https://www.youtube.com/playlist?list=PLrtCHHeadkHrIkSw0waA_S8p2G_lYxU_j
- https://www.youtube.com/playlist?list=PLBa-xZ_7YhjRBJJ7WpXZrdHEL05qlY5Uf
2. Product Teardowns
- https://www.youtube.com/playlist?list=PLHJHMiGyA5ULZij-SiHGt15x079Ty2mic

Podcasts

1. Real life experience of PM
- https://open.spotify.com/show/1XBrhVLsQOIAv3KFBqnzrX?si=d8c78b79a7fa47d9

Blogs

1. PM Questions (These 2 are from IIMB Seniors)
- https://technomanagers101.blogspot.com/
- https://naaavh.medium.com/
2. Upcoming Products
- https://www.producthunt.com/

Preparation

- https://www.linkedin.com/posts/shravantickoo_productmanagement-startups-growth-activity-6890991777747480576-k88x/

Ultimate Guides

- https://www.aha.io/roadmapping/guide/all
- https://github.com/ProductHired/open-product-management

Courses

1. Linked In – In case you have premium access
- https://www.linkedin.com/learning/becoming-a-product-manager-a-complete-guide
2. Udemy – Paid
- https://www.udemy.com/course/productmanagement101/

Mentoring

- https://www.seekamentor.com/
- https://www.productmanagementexercises.com/

Others

1. Collection of Newsletters
- https://www.linkedin.com/posts/shyveeshi_productmanagement-careers-newsletters-activity-6945031804139556864-QDjk/?

2. Collection of Groups
- https://thehiveindex.com/topics/product-management/
3. Conferences/ Webinars
- https://theproductmanager.com/topics/product-management-conferences/

Cheat Sheet

PRODUCT MANAGEMENT FRAMEWORKS

Category	Frameworks
Business/ strategy analysis	PEST, Business Model Canvas, SWOT, Porter's 5 forces, BCG Matrix, Kano, Ansoff Matrix, RACI/DACI, RFM, North star
Engagement Analysis	HEART, AARRR, AIDA, AARM, REAN, 5E's, HOOK canvas
Prioritization	MOSCOW, Business Value complexity, RICE, Eisenhower Method
Analyze	First Principles, 5 whys (5Ws & H), SPADE SCAMPER
Process	Design Thinking, CIRCLES, Double Diamond, DMAIC, Agile, Shape Up, MVP, GIST, Optimal product- process framework
Others	Dunning Kruger Effect, 4Ps, Crossing the Chasm, Toyota's way -14 principles.

The ones that are used often

Ideation -> White Boarding / Mindmaps
Macro view -> Business Canvas Model / SWOT
Solutioning -> Design Thinking
Quick Priority -> Value / Cost, RICE
Delivery -> Agile (Scrum)

PROBLEM SOLVING CHEATSHEET

THE PRODUCT FOLKS
Insurjo' 21

Problem-solving Framework
Understand & Define -> Solution & Validate -> Play & Prioritize -> Build, Launch & Iterate

Process:

1. Understand & review problems
2. Identity top variables
3. Weigh variables
4. Assumptions & constraints
5. Break it down & Prioritize
6. Validate solution
7. Plan / Timelines
8. Right-size MVP
9. Define success metrics
10. Iterate

Remember: Communicate at ALL TIMES (This is what we miss usually)
BVIR (Big visual Info Radiators)

Problem Analysis: Variables (cheatsheet)

Market / Ecosystem
-TAM/SAM/SOM
-Social/Political
-Regulatory/Compliance

Partners
-Impact/change reqd
-Scale changes

Competition
-Sustainable Competitive Advantage
-Competitive Differentiators

Customers
- Buyer, user, etc.
- Customer Value
- TCO
- Impact on existing experience

Internal
-Business/vision alignment
-Short-term & long-term goals
-Rev / Budget / ROI
-Stakeholders
-GTM
-End customer Experience
-KPIs for success
-Data
-Integrations

LEARNINGS FROM NEW BUSINESS/NEW TECH

Tech

1. People love new tech
2. Customer Value Creation
   Unlearn traditional ways + Identify new opportunities
3. Challenges far building good data/annotations
4. How your business and tech decouples - Be nimble
5. Ensure tech's value realization & usage.
6. Solve 1 primary use-case well.

Marketplace

1. While re-pivoting business, focus on the channel, delivery, customer value, incremental offerings, competition. Tech will be solved.
2. Look for Indirect competition
3. Customer trust & confidence.
4. Partners to scale
5. Find ways to build Network Effect.
6. Data & data derivations are pivotal.
7. Each customer needs equal care.
8. Focus on value than just sales nos.

Balanced Roadmaps

Delighters	Direct Requests	Big bets
Metric Movers	Sustenance	Quickwins (RoI)
Tech Backlog	Qualified Experiments	

PM Framework

Communication
- Empathy
- Clarity
- Stakeholder & cust. mgmnt

Execution
- Bias for action
- Drive & Passion
- Timelines
- Ownership
- Quality of work

Innovation
- What did you help identify, solve, create
- it's value to the org
- Risk-taking & mgmnt thereof

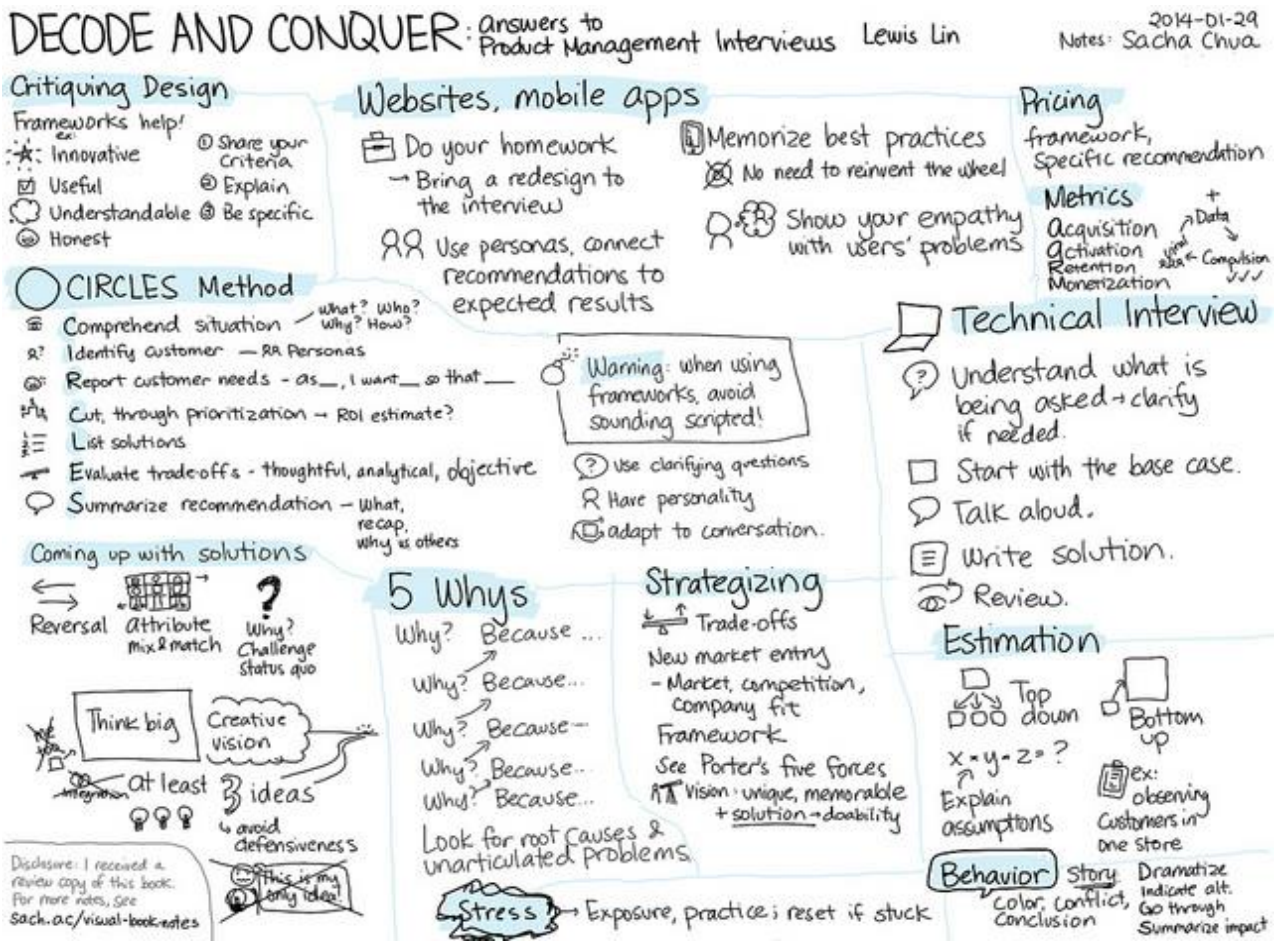
AS CUSTOMERS GROW (B2B GROWTH PHASES)

First 10 customers
- Find 1 good customer
- Be OPEN to CHANGE
- Hack & Agility
- Validation
- Evaluate PMF opportunities
- Focus on 1 CORE PROBLEM & solve it better than the rest.
- Prob Solver PMs(Generalists) 0-3
- Long term Roadmap 10-30%

10-100 customers
- Process Baselining
- Partner synergies
- Customer Success Focus
- Referencability
- Start thinking "self serve", if not already.
- 2+ Differentiators
- PMs : Process specific (3-10)
- Long term Roadmap (30% to 60%)

100+ Customers
- Repivot/Refocus, UX Revamp
- Multi geo/Muti-ntnl
- Defining & leading Industry
- Be closer to customers
- Playbooks / SOPs
- Good Data is pivotal
- PMs: Specialised (10+)
- Long term Roadmap (60% to 100%)

Fin!


DECODE AND CONQUER: Answers to Product Management Interviews Lewis Lin
2014-01-29
Notes: Sacha Chua
Critiquing Design
Frameworks help!
Innovative
Useful
Understandable
Honest
Share your criteria
Explain
Be specific
CIRCLES Method
Comprehend situation
Identify customer
Report customer needs
Cut, through prioritization
List solutions
Evaluate trade-offs
Summarize recommendation
Coming up with solutions
Reversal
Attribute mix & match
Why? Challenge status quo
Think big
Creative vision
At least 3 ideas
avoid defensiveness
Disclosure: I received a review copy of this book. For more notes, see sach.ac/visual-book-notes
Websites, mobile apps
Do your homework
Bring a redesign to the interview
Use personas, connect recommendations to expected results
Memorize best practices
No need to reinvent the wheel
Show your empathy with users' problems
Warning: when using frameworks, avoid sounding scripted!
Use clarifying questions
Have personality
adapt to conversation.
5 Whys
Why? Because ...
Look for root causes & unarticulated problems
Strategizing
Trade-offs
New market entry
- Market, competition, company fit
Framework
See Porter's five forces
Vision: unique, memorable + solution → doability
Stress
Exposure, practice; reset if stuck
Pricing
framework, Specific recommendation
Metrics
Acquisition
Activation
Retention
Monetization
Data
Technical Interview
Understand what is being asked → clarify if needed.
Start with the base case.
Talk aloud.
Write solution.
Review.
Estimation
Top down
Bottom up
x · y · z = ?
Explain assumptions
ex: observing customers in one store
Behavior
Story
Dramatize
Indicate alt.
Go through
Summarize impact
Gooooogle
4 billion searches/day
$60 billion annual rev.
30 trillion pages indexed
CPC: 2¢ - 10$ avg: 50¢
ad block 5% of users
50k employees
twitter
320M monthly active users
500M tweets/day
100M DAU
80% on mobile, 80% out of US
Instagram
300M users
40m pics/day
75m active daily
80 M users in US
files & file sizes
iPhone: 10 Mbps @ 720p ⇒ 80 MB/min
1 photo on phone ~ 2mb
1 tweet ~ 120 characters
1 unicode char ~ 1-15 bytes
10^0 → byte
10^3 → kb
10^6 → mb
10^9 → gb
10^12 → tb
vs.
US: 47% 47%
World: 70% 20%
China: 75% 20%
Smartphones
2 billion (30%) have world
60% of US has smartphones
12% of INDIA has smartphone
45% of china has smartphone
internet
3 billion internet users
75% of US has internet
40% of world has internet
facebook
1 billion active daily users
800M active daily Mobile users
85% active users outside US/CAN.
83% US internet users
50% of world web users
70% use daily
46min/day on Facebook + Insta.
1.5 billion searches/day
168 in minutes
People & the world
US - 300M
China - 1B
Europe - 700M
India - 1B
Latin A - 600M
World - 7B
avg city - 1M
NYC - 8M
Life Exp US - 80yr
Life Exp Wld - 66.7yr
avg 3/household
avg 2 bedroom - 800sqft
YouTube
48 hours/day
300 min Content uploaded
50% of viewing = mobile
Design ?s
clarify
prioritize/motivate
identify users
identify use cases
cur. products
improvements to?
wrapup/validate
product improvement
what is the goal
what problems
solutions
implementation
validate
Metrics
Acquisition - visiting site + initiating
Activation - happy 1st visit (views, x pages, y seconds/clicks email/blog signups, account signups)
Retention - email open, click through, repeat visit (3*/30days)
Referral - refer 1+ to visit, refer 1+ who activate
Revenue - user generates min rev./user generates breakeven
Google Photos
200M monthly AU
13.7 petabytes of photos
24 billion selfies
#numbers
~9000 hrs/year
~500,000 min/year
x% in 1 year years to double
avg apt = 1000 sq ft house = 2000 sq ft
Kids
90% of people eventually have kids ⇒ ~30% of life with kids ⇒ ~30% of households have kids
By: Noah Presler
noah@presler.me